STA1610 · INTRODUCTION TO STATISTICS

Exam Prep Revision Guide

Start here: recent paper evidence, exact question/page references, and independent step-by-step solutions for every topic.

23	topic chapters	**6**	calculator recipes
48	past-paper question entries	**2**	source papers with page references
23	worked solutions	**1**	complete formula sheet

This PDF is a print layout of the interactive study site. Every worked solution is shown in full — nothing is hidden behind a click. Past-paper references are exact; question wording is paraphrased and every solution is independently recomputed, so this is a revision aid, not an official memorandum.

How to use it: work the 23 chapters in order for coverage, or open the exam-prep chapter first and use the paper references to jump straight to weak topics. The formula sheet at the end is designed to be printed on its own for the final hour before the exam.

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Generated from the interactive study site (single HTML file). Print double-sided on A4; the contents page and PDF bookmarks are clickable.

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EXAM PREP

Past Papers & Worked Solutions

Start here: recent paper evidence, exact question/page references, and independent step-by-step solutions for every topic.

48	question entries inspected in the 2019 and 2024 sources
23	topic guides with a worked solution
3	honestly labelled exam-style gap fillers

How to use this section: attempt the paraphrased question first, open the worked solution second, then use “Revise this topic” for the theory. Use the source-page link to see the original paper, figure and answer options.

Verified recent paper library

2024 Oct/Nov assessment preview

23 questions · 12 pages · most recent usable paper found

Question text is screenshot-based. Some feedback is mismatched, so the solutions below are independently recomputed from the visible data.

https://www.studyhalo.com/media/documents/STA1610/2024/exam/STA1610-Exam_Oct-Nov_2024_Attempt_Review.pdf

https://www.studyhalo.com/modules/STA1610/

2019 May/June examination

25 questions · 22 pages · 2 hours · 100 marks

The clearest complete recent final-exam paper located. Questions occupy pages 2–11, followed by formulae and statistical tables.

https://www.studyhalo.com/media/documents/STA1610/2019/exam/STA1610-2019-6-E-1.pdf

https://www.studyhalo.com/modules/STA1610/

2-hour practice timer

Two-hour practice timer — available in the interactive version of this guide (it cannot run on paper: use a clock and the 25-question structure above).

Year	What was found	How it is used
2019	Complete May/June final paper	Used for question references and solutions.
2020	No public final paper verified in this search	Not filled with mislabelled material.
2021	Oct/Nov mock listing; only a short public preview	Not used for full question extraction.
2022	No public final paper verified	Assignments are not relabelled as exams.
2023	Assessments found; no public final paper verified	Not treated as a final-exam paper.
2024	Oct/Nov assessment preview	Used, with visible-data checks and source warnings.
2025	Commercial “exam pack” titles, but inspected listings contain older papers	No unverified 2025 final paper claimed.

A StudyHalo file labelled “2019 Oct/Nov” was also excluded after its contents were found to be an Oct/Nov 2018 paper. “Not verified” means not located publicly in this search—not that a paper does not exist in myUnisa.

Past-paper sections by topic

Paper references are exact; questions are paraphrased for revision. Worked solutions are independently written, not official memoranda.

01 What is Statistics? [PAST-PAPER BACKED]

`2019 May/Jun · Q8 · p.5 ↗ source`

> **Question (paraphrased):** Which of μ, s, $\bar{x}$ and σ are population parameters rather than sample statistics?

Related: 2021 mock Q1 also tests the meaning of a statistic.

WORKED SOLUTION

1. μ is the population mean and σ is the population standard deviation, so both are parameters.
2. $\bar{x}$ is the sample mean and s is the sample standard deviation, so both are statistics.

> Answer: A and D: μ and σ (option 2).

Exam tip: Greek letters usually denote population parameters; $\bar{x}$ and s denote sample statistics.

02 Types of Data [PAST-PAPER BACKED]

`2024 Oct/Nov · Q1 · p.1 ↗ source`

> **Question (paraphrased):** Classify education level recorded as Primary / Secondary / Higher, and the number of people with disabilities in a team.

Also tested in 2019 Q1–Q2, p.2.

WORKED SOLUTION

1. Education level has categories with a meaningful order, but the gaps between categories are not numerical measurements: qualitative ordinal.
2. A number of people is obtained by counting (0, 1, 2, …): quantitative discrete.

> Answer: Education level: qualitative ordinal. Number of people: quantitative discrete.

Exam tip: Nominal = labels; ordinal = ordered labels; discrete = counts; continuous = measurements.

03 Sampling Methods [EXAM-STYLE GAP]

`Exam-style gap filler · no direct question located`

Question (paraphrased): A university divides students into faculties and then randomly selects students from every faculty. Which sampling method is this?

Source / scope note: No direct sampling-method question was found in the located 2019 or 2024 papers. This is an original exam-style example, not a past-paper question.

WORKED SOLUTION

1. Faculties form subgroups (strata) covering the population.
2. Random selection then takes place within every stratum. This is the defining feature of stratified random sampling.

Answer: Stratified random sampling.

Exam tip: Stratified: sample within every subgroup. Cluster: randomly select whole clusters, then study members of the chosen clusters.

04 Frequency Distributions & Tables [PAST-PAPER BACKED]

2024 Oct/Nov · Q3 · p.2 ↗ source

Question (paraphrased): For classes 10–19, 20–29, 30–39, 40–49, 50–59 with frequencies 10, 14, 17, 7, 2, find the number of classes and the percentage of observations ≤ 49.

Also tested in 2019 Q3–Q4, p.3.

Source / scope note: The review screenshot's printed feedback refers to a different variant (class width and ≤ 39). This solution is recomputed from the visible table.

WORKED SOLUTION

1. There are 5 classes and n = 10 + 14 + 17 + 7 + 2 = 50.
2. Cumulative frequency through 49 = 10 + 14 + 17 + 7 = 48.
3. Percentage ≤ 49 = (48 / 50) × 100 = 96%.

Answer: 5 classes; 96% of observations are ≤ 49.

Exam tip: Do not confuse number of classes (5), class width (10) and cumulative percentage (96%).

05 Measures of Central Tendency [PAST-PAPER BACKED]

2024 Oct/Nov · Q4 · p.3 ↗ source

Question (paraphrased): Calculate the mean of 12, 14, 11, 16, 9, 8, 7, 6, 20, 12, 8, 13, 12, 7 days.

Also tested in 2019 Q5–Q6, p.4.

WORKED SOLUTION

1. There are n = 14 observations and $\Sigma x = 155$.
2. $\bar{x} = \Sigma x / n = 155 / 14 = 11.0714\ldots$

Answer: Mean = 11.1 days (to one decimal place).

Exam tip: 2019 Q5 reinforces that an extreme outlier strongly affects the mean, while the median is much more resistant.

06 Measures of Dispersion [PAST-PAPER BACKED]

2024 Oct/Nov · Q6 · p.4 ↗ source

Question (paraphrased): Calculate the sample standard deviation of 117, 105, 120, 107, 120, 124.

WORKED SOLUTION

1. n = 6 and $\bar{x} = 693 / 6 = 115.5$.
2. $\Sigma(x - \bar{x})^2 = 297.5$.

3 For a sample, $s^2 = 297.5 / (6 - 1) = 59.5$.

4 $s = \sqrt{59.5} = 7.7136...$

> Answer: Sample standard deviation = 7.71 (option c).

Exam tip: Sample variance divides by n − 1. Here 59.50 is the variance; 7.71 is the standard deviation.

07 Measures of Position & Box Plots [PAST-PAPER BACKED]

2019 May/Jun · Q7 · p.4 ↗ source

> **Question (paraphrased):** For the ordered data 1, 2, 3, 4, 5, 6, use the module's position method to evaluate Q3.

Also tested in 2024 Q5, p.3.

WORKED SOLUTION

1 Using $L_p = (n + 1)p$, the Q3 position is $7 \times 0.75 = 5.25$.

2 Move 0.25 of the way from the 5th value (5) to the 6th value (6): $Q3 = 5 + 0.25(6 - 5) = 5.25$.

3 For comparison, Q2 = 3.5, Q1 = 1.75 and IQR = 5.25 − 1.75 = 3.50.

> Answer: Q3 = 5.25 (option 4).

Exam tip: Quartile conventions differ between software packages. In the exam, use the position/interpolation convention taught in STA1610.

08 Probability Basics [PAST-PAPER BACKED]

2024 Oct/Nov · Q7 · p.5 ↗ source

> **Question (paraphrased):** Using the visa table, find P(Lesotho or Other), given N = 2,004,425; Lesotho total = 227,770; Other total = 422,768; Lesotho ∩ Other = 75,164.

WORKED SOLUTION

1 Use the addition rule: $P(L \cup O) = P(L) + P(O) - P(L \cap O)$.

2 Favourable count = 227,770 + 422,768 − 75,164 = 575,374.

3 $P(L \cup O) = 575,374 / 2,004,425 = 0.2870519...$

> Answer: P(Lesotho or Other) = 0.2871.

Exam tip: Subtract the overlap once; otherwise the Lesotho–Other cell is counted twice.

09 Probability Rules [PAST-PAPER BACKED]

2019 May/Jun · Q9 · p.5 ↗ source

> **Question (paraphrased):** If P(A) = 0.15 and P(B) = 0.35, assess the claim that independence implies P(A or B) = 0.5525.

Conditional probability is also tested in 2024 Q8, p.6.

WORKED SOLUTION

1 Under independence, $P(A \cap B) = P(A)P(B) = 0.15 \times 0.35 = 0.0525$.

2 Therefore $P(A \cup B) = 0.15 + 0.35 - 0.0525 = 0.4475$.

3 The claimed 0.5525 is wrong; it adds the intersection instead of subtracting it.

> Answer: The independent-union statement is incorrect; the correct probability is 0.4475.

Exam tip: Independent does not mean mutually exclusive. Independent: multiply for AND. Mutually exclusive: intersection is zero.

10 Bayes' Theorem [EXAM-STYLE GAP]

Exam-style gap filler · related skill in 2024 Q8

Question (paraphrased): Supplier A provides 60% of components and B provides 40%. Their defect rates are 2% and 5%. A chosen component is defective. What is P(B | Defective)?

Source / scope note: 2024 Q8 tests conditional probability, but no full Bayes reversal was found in the two located papers. This is an original exam-style example.

WORKED SOLUTION

1. Total probability: P(D) = 0.60(0.02) + 0.40(0.05) = 0.012 + 0.020 = 0.032.
2. Bayes' theorem: P(B | D) = P(B)P(D | B) / P(D) = 0.020 / 0.032 = 0.625.

Answer: P(B | Defective) = 0.625 = 62.5%.

Exam tip: First calculate the total probability of the observed evidence; then take the relevant branch divided by that total.

11 Discrete Random Variables [PAST-PAPER BACKED]

2024 Oct/Nov · Q9 · p.7 ↗ source

Question (paraphrased): For X = 0, 1, 2, 3, 4 with probabilities 0.3, 0.2, 0.2, 0.2, 0.1, calculate E[X].

Added from the paper audit: this exam topic was missing from the original site. Also see 2019 Q11, p.6.

WORKED SOLUTION

1. Use $E[X] = \Sigma x P(x)$.
2. E[X] = 0(0.3) + 1(0.2) + 2(0.2) + 3(0.2) + 4(0.1).
3. E[X] = 0 + 0.2 + 0.4 + 0.6 + 0.4 = 1.6.

Answer: Expected value E[X] = 1.60.

Exam tip: Expected value need not be one of the possible x-values. Check first that the probabilities sum to 1.

12 The Binomial Distribution [PAST-PAPER BACKED]

2024 Oct/Nov · Q10 · p.7 ↗ source

Question (paraphrased): In a binomial experiment with n = 7 and success probability π = 0.15, find P(X ≥ 1).

Binomial variance is also tested in 2019 Q12, p.6.

WORKED SOLUTION

1. Use the complement: P(X ≥ 1) = 1 − P(X = 0).
2. $P(X = 0) = (1 - 0.15)^7 = 0.85^7 = 0.3205771\ldots$
3. P(X ≥ 1) = 1 − 0.3205771… = 0.6794229…

Answer: P(X ≥ 1) = 0.6794.

Exam tip: "At least one" is usually fastest as 1 − P(none).

13 The Poisson Distribution [PAST-PAPER BACKED]

2024 Oct/Nov · Q11 · p.8 ↗ source

Question (paraphrased): Late arrivals average λ = 2.5 managers per day. Find the probability of exactly one late arrival on a day.

A cumulative Poisson probability is tested in 2019 Q13, p.7.

WORKED SOLUTION

1. Use $P(X = x) = e^{-\lambda} \lambda^x / x!$ with x = 1.
2. $P(X = 1) = e^{-2.5} \times 2.5 / 1! = 0.2052125\ldots$

Answer: $P(X = 1) = 0.2052$.

Exam tip: Poisson models a count over a fixed interval. Read carefully: exactly one and at least one require different calculations.

14 The Normal Distribution [PAST-PAPER BACKED]

2024 Oct/Nov · Q14 · p.9 ↗ source

Question (paraphrased): Tourist stay is $X \sim N(14, 3^2)$. Find x such that $P(X \geq x) = 0.40$.

Also see 2024 Q12–Q13, p.8 and 2019 Q14–Q15, p.7.

WORKED SOLUTION

1. A right-tail area of 0.40 means the cumulative area to the left is 0.60.
2. From the z-table, $z \approx 0.255$ for cumulative area 0.60.
3. Back-transform: $x = \mu + z\sigma = 14 + 0.255(3) = 14.765$.

Answer: $x = 14.765$ days.

Exam tip: For "at least x", use a right tail. With unrounded tables $z = 0.2533$ gives $x \approx 14.760$; the paper's answer uses $z = 0.255$.

15 Sampling Distributions & the CLT [PAST-PAPER BACKED]

2024 Oct/Nov · Q15 · p.9 ↗ source

Question (paraphrased): A sample of $n = 20$ comes from a population with standard deviation $\sigma = 12$. Determine the standard error of the mean.

Also see 2024 Q16 (proportion SE), p.9 and 2019 Q16–Q17, p.8.

WORKED SOLUTION

1. The standard error of $\bar{x}$ is $\sigma_{\bar{x}} = \sigma / \sqrt{n}$.
2. $\sigma_{\bar{x}} = 12 / \sqrt{20} = 2.68328...$

Answer: Standard error = 2.68.

Exam tip: The population mean (100) is not used in the standard-error calculation. Do not confuse σ with $\sigma/\sqrt{n}$.

16 Confidence Intervals [PAST-PAPER BACKED]

2024 Oct/Nov · Q17 · p.9 ↗ source

Question (paraphrased): For $n = 30$, $\bar{x} = 110$ and sample standard deviation $s = 21$, construct a 90% confidence interval for μ.

Also see 2024 Q18 and 2019 Q19–Q20, p.9.

WORKED SOLUTION

1. σ is unknown, so use t with $df = n - 1 = 29$. For 90% confidence, $t_{0.05,29} \approx 1.699$.
2. $SE = s / \sqrt{n} = 21 / \sqrt{30} = 3.83406...$
3. Margin $= 1.699(3.83406...) = 6.5141...$
4. CI $= 110 \pm 6.5141... = (103.486, 116.514)$.

Answer: 90% CI ≈ (103.49; 116.51).

Exam tip: Because s (not σ) is supplied, the t-interval gives the listed answer; using $z = 1.645$ gives a narrower, incorrect interval here.

17 Hypothesis Testing Framework [PAST-PAPER BACKED]

2019 May/Jun · Q22 · p.10 ↗ source

Question (paraphrased): A one-tailed test has p-value 0.0092. What is the p-value for the corresponding two-tailed test?

A full critical-value and p-value calculation appears in 2024 Q19, p.10.

WORKED SOLUTION

1. For the corresponding two-sided test on a symmetric test distribution, include both equal-tail areas.
2. Two-tailed p = 2 × one-tailed p = 2(0.0092) = 0.0184.

Answer: Two-tailed p-value = 0.0184 (option 3).

Exam tip: Then compare the p-value with α: reject H_0 when $p \le \alpha$.

18 Z-Test for the Population Mean [PAST-PAPER BACKED]

2024 Oct/Nov · Q19 · p.10 ↗ source

Question (paraphrased): Test H_0: $\mu \le 14$ versus H_1: $\mu > 14$ at $\alpha = 0.05$, using $n = 20$, $\bar{x} = 15.6$ and known $\sigma = 4.4$.

WORKED SOLUTION

1. SE = 4.4 / √20 = 0.98387...
2. z = (15.6 − 14) / 0.98387... = 1.626.
3. Right-tail critical value = 1.645; p-value = P(Z ≥ 1.626) ≈ 0.0520.
4. Because 1.626 < 1.645 (equivalently 0.0520 > 0.05), do not reject H_0.

Answer: z = 1.626; critical value = 1.645; p ≈ 0.052. Do not reject H_0.

Exam tip: Use σ/√n, not σ/n. This item is close to the 5% boundary, so avoid premature rounding.

19 T-Test for the Population Mean [EXAM-STYLE GAP]

Exam-style gap filler · related t-methods in 2019 Q18 and 2024 Q17

Question (paraphrased): For H_0: $\mu = 50$ versus H_1: $\mu \ne 50$, a sample gives $n = 16$, $\bar{x} = 45.6$ and $s = 8$. Test at $\alpha = 0.05$.

Source / scope note: The located papers test t-confidence intervals and when to use t-procedures, but no complete numerical t-test was found. This is an original exam-style example.

WORKED SOLUTION

1. σ is unknown, so $t = (\bar{x} - \mu_0) / (s/\sqrt{n}) = (45.6 - 50) / (8/4) = -2.20$.
2. Degrees of freedom = 15; two-tailed critical values are ±2.131.
3. Because $|-2.20| = 2.20 > 2.131$, reject H_0. The two-tailed p-value is approximately 0.0439.

Answer: t = −2.20, df = 15, p ≈ 0.0439. Reject H_0 at 5%.

Exam tip: Use t when σ is unknown; the degrees of freedom and one- versus two-tailed alternative determine the critical region.

20 Z-Test for a Population Proportion [PAST-PAPER BACKED]

2019 May/Jun · Q21 · p.9 ↗ source

Question (paraphrased): A magazine claims that more than 40% of households changed lifestyle. In n = 100, 67 changed. Test the claim at α = 0.10.

Source / scope note: Source warning: as printed, option 1 has the wrong alternative tail and option 4 says p = 0.10. Both appear incorrect, so the MCQ lacks a unique correct option. The full correct test is shown here.

WORKED SOLUTION

1. H_0: $p = 0.40$ versus H_1: $p > 0.40$; $\hat{p} = 67/100 = 0.67$.

2 Under H_0, SE = $\sqrt{[0.40(0.60)/100]} = 0.0489898...$

3 $z = (0.67 - 0.40) / 0.0489898... = 5.511$.

4 The 10% right-tail critical value is 1.28 and the p-value is below 0.0001, so reject H_0.

> Answer: z = 5.51; reject H_0. There is very strong evidence that $p > 0.40$.

Exam tip: For a proportion hypothesis test, calculate the standard error using the null value p_0, not $\hat{p}$.

21 Chi-Square Tests [PAST-PAPER BACKED]

2024 Oct/Nov · Q20–Q21 · pp.10–11 ↗ source

Question (paraphrased): A 5 × 4 contingency table is tested for independence at $\alpha = 0.05$. If $\chi^2 = 20.11$, find df, the critical value and the decision.

Also tested in 2019 Q23–Q24, pp.10–11.

WORKED SOLUTION

1 $df = (r - 1)(c - 1) = (5 - 1)(4 - 1) = 12$.

2 The upper-tail 5% critical value is $\chi^2_{0.05,12} = 21.026$.

3 Since $20.11 < 21.026$, do not reject H_0. There is insufficient evidence of dependence.

> Answer: df = 12; critical value = 21.026; do not reject independence.

Exam tip: Independence is H_0; dependence/association is H_1. Reject only when the calculated χ^2 exceeds the upper-tail critical value.

22 Correlation [PAST-PAPER BACKED]

2024 Oct/Nov · Q22 · p.11 ↗ source

Question (paraphrased): The estimated line is $\hat{y} = -2x + 7$ and $R^2 = 0.90$. Find the correlation coefficient r.

WORKED SOLUTION

1 $|r| = \sqrt{R^2} = \sqrt{0.90} = 0.94868...$

2 The slope is negative (−2), so r must also be negative.

3 $r = -0.94868... \approx -0.95$.

> Answer: r = −0.95.

Exam tip: R^2 has no sign. Take the sign of r from the slope of the regression line.

23 Simple Linear Regression [PAST-PAPER BACKED]

2024 Oct/Nov · Q23 · p.12 ↗ source

Question (paraphrased): Least-squares output gives $SSE = \Sigma(y_i - \hat{y}_i)^2 = 9{,}314.22$ and $SST = \Sigma(y_i - \bar{y})^2 = 24{,}511.11$. Calculate R^2.

Regression-equation interpretation is also tested in 2019 Q25, p.11.

WORKED SOLUTION

1 Use $R^2 = 1 - SSE/SST$.

2 $R^2 = 1 - 9{,}314.22 / 24{,}511.11 = 1 - 0.380000... = 0.620000...$

> Answer: $R^2 = 0.62$; about 62% of the variation in y is explained by the model.

Exam tip: $SSR = SST - SSE$ and $R^2 = SSR/SST = 1 - SSE/SST$.

TOPIC 01

What is Statistics?

The science of collecting, organising, analysing, and interpreting data to make decisions

Core idea. Statistics is the set of methods we use to turn raw numbers into meaningful conclusions. Whenever you see a headline like "72% of South Africans support renewable energy" or "the average matric pass rate increased by 3%," someone used statistics to arrive at that number.

1.1 Two Branches of Statistics

Descriptive Statistics

Descriptive statistics *summarise* and *describe* the data you have collected. You are not trying to generalise beyond your data — you are simply painting a picture of what the data looks like.

Examples: calculating the average exam mark of your class, drawing a bar chart of favourite colours, finding the percentage of students who passed.

Inferential Statistics

Inferential statistics use a *sample* (a small group) to make conclusions about a *population* (the entire group you care about). This is where probability enters the picture, because you are always dealing with uncertainty when you generalise from a sample.

Examples: polling 1 000 voters to predict an election outcome, testing 50 light bulbs to estimate the lifespan of all bulbs produced by a factory.

Analogy. Descriptive statistics is like taking a photograph of a crowd — you capture exactly what is there. Inferential statistics is like looking at a small section of the crowd through binoculars and trying to guess what the entire crowd looks like.

1.2 Key Vocabulary

Term	Definition	Example
Population	The entire group of individuals or items you want to study	All 40 000 UNISA students enrolled in 2025
Sample	A subset of the population that you actually collect data from	500 randomly selected UNISA students
Parameter	A numerical summary of the population (usually unknown)	The true average age of all UNISA students (μ)
Statistic	A numerical summary of the sample (calculated from your data)	The average age of the 500 surveyed students ($\bar{x}$)
Variable	A characteristic that can take different values for different individuals	Age, height, exam mark, favourite colour
Data	The actual observed values of a variable	22, 25, 19, 31, … (ages of surveyed students)

Exam tip. STA1610 exams frequently test the distinction between population vs sample and parameter vs statistic. A simple trick: Population → Parameter; Sample → Statistic. If the question says "all" or "every," it is a population/parameter. If it says "selected," "surveyed," or "sample of," it is a sample/statistic.

1.3 The Statistical Process

1. **Identify the problem.** What question do you want to answer? (e.g., "Do UNISA students who attend tutorials score higher?")
2. **Collect data.** Design a study and gather observations (e.g., survey 200 students about tutorial attendance and exam marks).
3. **Organise and summarise.** Use tables, graphs, and descriptive statistics (e.g., calculate the mean mark for attendees vs non-attendees).
4. **Analyse and interpret.** Use inferential statistics to draw conclusions (e.g., perform a hypothesis test to determine if the difference in means is statistically significant).
5. **Make a decision.** Act on your findings (e.g., recommend that the university invest in more tutorial sessions).

1.4 Why Study Statistics?

- **Evidence-based decisions:** Statistics prevents you from relying on gut feelings or anecdotes. Instead of saying "I think most students prefer online classes," you can say "78% of a random sample of 1 000 students prefer online classes, with a margin of error of 3%."
- **Detecting patterns:** Human brains are bad at seeing patterns in large data sets. Statistics provides tools to reveal trends, relationships, and outliers that would otherwise be invisible.
- **Understanding research:** Every scientific paper, news article, and government report uses statistics. Without statistical literacy, you cannot critically evaluate claims like "this medicine reduces risk by 40%."
- **Career relevance:** Statistics is used in business, medicine, psychology, engineering, finance, marketing, and virtually every field that deals with data.

[CONCEPT] Identifying population, sample, parameter, statistic

A researcher wants to know the average monthly income of all small business owners in Gauteng. She randomly selects 300 small business owners and finds their average monthly income is R42 500. Identify the population, sample, parameter, and statistic.

Step 1 — Identify the population.

> The population is the entire group the researcher wants to learn about: *all small business owners in Gauteng*.

Step 2 — Identify the sample.

> The sample is the group she actually collected data from: *the 300 randomly selected small business owners*.

Step 3 — Identify the parameter.

> The parameter is the numerical summary of the population: *the true average monthly income of all small business owners in Gauteng* (this is unknown — it is what she is trying to estimate).

Step 4 — Identify the statistic.

> The statistic is the numerical summary of the sample: *R42 500* (the average monthly income of the 300 surveyed owners).

> Population: all small business owners in Gauteng. Sample: 300 selected owners. Parameter: true mean income (unknown). Statistic: R42 500.

TOPIC 02

Types of Data

Classifying variables so you can choose the right analysis method

> Core idea. Not all data is the same. The type of data you have determines which statistical methods are appropriate. Using the wrong method for your data type leads to meaningless results. The first step in any statistical analysis is to classify your variables correctly.

2.1 The Data Classification Tree

Every variable falls into one of two broad categories: **qualitative** (categorical) or **quantitative** (numerical). Quantitative variables are further split into **discrete** and **continuous**.

Qualitative (Categorical) Data

Qualitative data describes *qualities* or *categories*. The values are labels, not numbers you can meaningfully add or average.

Examples: gender (male/female), eye colour (brown/blue/green), degree type (BCom/BSc/BA), satisfaction level (low/medium/high).

Qualitative data is further divided into:

- **Nominal:** Categories with no natural order. You cannot say one category is "more" or "less" than another. Examples: gender, race, favourite colour, province.
- **Ordinal:** Categories with a meaningful order, but the distances between categories are not equal or measurable. Examples: education level (matric < diploma < degree < masters), satisfaction (dissatisfied < neutral < satisfied), class rank (1st, 2nd, 3rd).

Quantitative (Numerical) Data

Quantitative data consists of *numbers* that represent measurable quantities. You can meaningfully add, subtract, and average these values.

Examples: age, height, weight, exam mark, income, number of children.

Quantitative data is further divided into:

- **Discrete:** Values that can only take specific, separate values — usually whole numbers. You can *count* discrete data. Examples: number of siblings (0, 1, 2, 3…), number of cars in a parking lot, number of goals scored in a match.
- **Continuous:** Values that can take any value within a range, including decimals and fractions. You *measure* continuous data. Examples: height (172.3 cm), weight (68.45 kg), time (3.72 seconds), temperature (22.8°C).

How to tell discrete from continuous. Ask yourself: "Can this value take on a fraction?" If you are counting people, the answer is no — you can't have 2.5 people. That's discrete. If you are measuring height, the answer is yes — someone can be 172.3 cm or 172.31 cm. That's continuous. A useful rule of thumb: counted data is discrete; measured data is continuous.

2.2 Levels of Measurement

A more detailed classification system assigns each variable to one of four **levels of measurement**. Each level includes all the properties of the levels below it:

Level	Properties	Examples	Operations Allowed
Nominal	Categories only; no order	Gender, blood type, car brand	Count, mode
Ordinal	Categories with order; unequal gaps	Letter grades (A, B, C), Likert scale	Count, mode, median
Interval	Ordered; equal gaps; no true zero	Temperature in °C, IQ score, calendar year	+, −, mean, std dev
Ratio	Ordered; equal gaps; true zero exists	Height, weight, income, age, distance	+, −, ×, ÷, all statistics

What is a "true zero"? A true zero means that a value of zero represents the complete absence of the quantity. For weight, 0 kg means no weight at all — that's a true zero (ratio level). For temperature in °C, 0°C does not mean "no temperature" — it's just the freezing point of water. You can have negative temperatures. So °C is interval, not ratio. But temperature in Kelvin does have a true zero (0 K = absolute zero = no molecular motion), so Kelvin is ratio.

[EXAM-STYLE] Classifying variables

Classify each of the following variables by type (qualitative/quantitative) and level of measurement:

(a) The brand of cellphone a student owns (Samsung, Apple, Huawei, etc.)

Type: Qualitative (categorical labels, not numbers).

Level: Nominal (no natural order — Samsung is not "more" than Apple).

(b) The number of modules a student is registered for

Type: Quantitative, discrete (you count modules: 3, 4, 5…).

Level: Ratio (0 modules means no modules — true zero exists).

(c) The temperature in Johannesburg in °C

Type: Quantitative, continuous (temperature can be 22.7°C).

Level: Interval (equal gaps between degrees, but 0°C is not "no temperature").

(d) Student satisfaction: Very Dissatisfied, Dissatisfied, Neutral, Satisfied, Very Satisfied

Type: Qualitative (categories, not numbers).

Level: Ordinal (there is a clear order from dissatisfied to satisfied, but the "distance" between categories is not measurable).

(e) The time (in minutes) it takes a student to complete an exam

Type: Quantitative, continuous (time can be 87.3 minutes).

Level: Ratio (0 minutes means no time — true zero).

(a) Qualitative, Nominal. (b) Quantitative discrete, Ratio. (c) Quantitative continuous, Interval. (d) Qualitative, Ordinal. (e) Quantitative continuous, Ratio.

Exam tip. STA1610 exams almost always include a question asking you to classify 4–6 variables. The most common trap is confusing interval and ratio. Ask yourself: "Does zero mean nothing?" If yes → ratio. If no → interval. Another common trap: "number of..." variables are always discrete, even if the numbers are large. "Amount of..." or "time to..." variables are usually continuous.

TOPIC 03

Sampling Methods

How to select a representative subset from a population

Core idea. A census collects data from every member of the population — ideal but usually impossible due to cost, time, and logistics. A sample collects data from a subset. The quality of your statistical conclusions depends entirely on how well your sample represents the population. A biased sample leads to biased conclusions, no matter how sophisticated your analysis.

Analogy. Imagine you want to know if a pot of soup is salty enough. You don't need to drink the entire pot (census). You stir it well and taste one spoonful (sample). If the soup is well stirred (random sampling), the spoonful represents the whole pot. If you only taste from the top where the salt hasn't dissolved yet (biased sampling), your conclusion will be wrong.

3.1 Probability Sampling Methods

In probability sampling, every member of the population has a *known, non-zero chance* of being selected. This is the gold standard because it minimises bias and allows you to calculate margins of error.

3.1.1 Simple Random Sampling (SRS)

Every member of the population has an *equal* chance of being selected, and every possible sample of size n is equally likely.

How to do it: Assign a number to every member of the population. Use a random number generator (or draw names from a hat) to select n numbers.

Example: You have a list of 5 000 UNISA students. You use a random number generator to pick 200 unique numbers between 1 and 5 000. Those 200 students are your sample.

Advantage: Simple, unbiased, easy to understand.

Disadvantage: Requires a complete list of the population (a "sampling frame"), which is often unavailable. Can be expensive if the population is geographically spread out.

3.1.2 Systematic Sampling

Select every k-th member from a list after a random starting point.

How to do it:

1. Calculate the sampling interval: k = N / n, where N is the population size and n is the desired sample size.
2. Randomly select a starting point between 1 and k.
3. Select every k-th member from that starting point.

[CONCEPT] Systematic sampling

You want to select 50 students from a list of 1 000. Describe the systematic sampling procedure.

Step 1 — Calculate k.

> k = N / n = 1000 / 50 = 20

> You will select every 20th student.

Step 2 — Choose a random starting point.

> Randomly pick a number between 1 and 20. Suppose you get 7.

Step 3 — Select every 20th student.

> Your sample includes students at positions: 7, 27, 47, 67, 87, 107, ..., 987.

> 50 students selected systematically from the list of 1 000.

Advantage: Easier to implement than SRS, especially with physical lists.

Disadvantage: If the list has a hidden periodic pattern that matches k, the sample can be severely biased. For example, if every 20th house on a street is a corner house, and you sample every 20th house, you'll only get corner houses.

3.1.3 Stratified Sampling

Divide the population into homogeneous subgroups called **strata** (e.g., by gender, age group, faculty), then take a simple random sample from *each* stratum.

Why use it: Ensures that every important subgroup is represented in the sample in proportion to its size in the population. This reduces sampling error compared to SRS.

[CONCEPT] Stratified sampling

A university has 6 000 students: 3 600 undergraduates and 2 400 postgraduates. You want a stratified sample of 300 students proportional to each group. How many from each stratum?

Step 1 — Calculate the proportion of each stratum.

> Undergraduates: 3600 / 6000 = 0.60 (60%)

> Postgraduates: 2400 / 6000 = 0.40 (40%)

Step 2 — Multiply each proportion by the sample size.

> Undergraduates: 0.60 × 300 = 180

Postgraduates: 0.40 × 300 = 120

Select 180 undergraduates and 120 postgraduates using simple random sampling within each group.

3.1.4 Cluster Sampling

Divide the population into naturally occurring groups called **clusters** (e.g., schools, city blocks, departments). Randomly select a few *entire* clusters and survey *everyone* within those clusters.

When to use it: When the population is geographically spread out and it would be too expensive to sample individuals from everywhere. Cluster sampling saves travel costs.

Difference from stratified: In stratified sampling, you sample *some from every* group. In cluster sampling, you sample *all from some* groups.

[CONCEPT] Cluster sampling

A researcher wants to survey high school learners across Gauteng. There are 800 high schools. She randomly selects 20 schools and surveys every learner in those 20 schools. What sampling method is this?

Identify the method.

The population is divided into naturally occurring groups (schools = clusters).

A random selection of entire clusters is made (20 out of 800 schools).

Every member of the selected clusters is surveyed.

This is cluster sampling. The clusters are the schools.

3.2 Non-Probability Sampling Methods

In non-probability sampling, not every member has a known chance of selection. These methods are cheaper and faster but introduce bias, so you cannot generalise results to the population with statistical confidence.

3.2.1 Convenience Sampling

Select whoever is easiest to reach.

Example: Standing outside the library and surveying the first 50 students who walk past.

Problem: Students who use the library may be more studious than average, so your sample is biased toward higher-performing students.

3.2.2 Judgement (Purposive) Sampling

The researcher uses their own judgement to select "typical" or "informative" participants.

Example: A journalist interviewing only experts they believe represent the consensus view on climate change.

Problem: The researcher's biases directly influence who is included.

3.2.3 Quota Sampling

Similar to stratified sampling, but participants within each subgroup are selected by convenience rather than randomly.

Example: A market researcher needs 50 men and 50 women. She stands at a mall and surveys the first 50 men and 50 women she encounters.

Problem: The proportions are correct, but the individuals within each group are not randomly selected.

3.2.4 Snowball Sampling

Existing participants recruit new participants from their social networks. Used for hard-to-reach populations.

Example: Studying undocumented immigrants — you find one participant, who introduces you to two friends, who each introduce you to more.

Problem: The sample is limited to people within the same social network, which may not represent the broader population.

3.3 Sampling Errors and Bias

Type	Definition	Can it be eliminated?
Sampling error	The natural difference between a sample statistic and the population parameter due to random chance	No, but it decreases as n increases
Non-sampling error (bias)	Systematic errors from poor study design, non-response, measurement errors, or non-random sampling	Yes, through careful design

Common exam mistake. Students often confuse sampling error with bias. Sampling error is random and unavoidable — it's the reason we report margins of error. Bias is systematic and avoidable — it's the reason we use random sampling methods. A large sample reduces sampling error but does not fix bias. A biased sample of 1 million people is still biased.

Exam tip. STA1610 exams typically ask you to: (a) identify the sampling method used in a scenario, (b) explain the difference between probability and non-probability sampling, (c) calculate stratum sizes for stratified sampling, and (d) identify potential sources of bias. Memorise the four probability methods (SRS, systematic, stratified, cluster) and the four non-probability methods (convenience, judgement, quota, snowball). The key distinguishing feature of stratified vs cluster is: stratified = some from every group; cluster = all from some groups.

TOPIC 04

Frequency Distributions & Tables

Organising raw data so patterns become visible

4.1 Why Do We Need Frequency Tables?

Imagine your lecturer hands you a list of 200 exam scores written in no particular order. Staring at that wall of numbers tells you almost nothing. You cannot quickly see what the most common score was, how many students passed, or whether most scores clustered around 50 or around 70.

A **frequency table** solves this problem. It takes messy, unorganised data and sorts it into neat categories, then counts how many data values fall into each category. That count is called the **frequency**.

> Think of it like sorting a bag of mixed sweets. Instead of staring at a jumbled pile, you separate them by colour and count: 12 red, 8 green, 15 yellow. Now you instantly know which colour is most common. A frequency table does exactly this with numbers.

4.2 Ungrouped Frequency Tables (Discrete Data)

When the data values are individual numbers (usually whole numbers that repeat), we list each unique value and count how many times it appears.

Key Definitions

- **Frequency (f):** The number of times a particular value occurs in the dataset.
- **Relative Frequency:** The frequency divided by the total number of observations. It tells you the proportion (or percentage) of the data that falls in that category.
- **Cumulative Frequency:** A running total of frequencies. You add each frequency to the sum of all frequencies before it.

> Relative Frequency Formula
> Relative Frequency = f / n where f = frequency of the value and n = total number of observations. The sum of all relative frequencies must equal 1 (or 100%).

[CONCEPT] Building an ungrouped frequency table

Twenty students were asked how many siblings they have. The responses were:
2, 1, 0, 3, 1, 2, 4, 1, 0, 2, 1, 3, 2, 1, 0, 2, 5, 1, 3, 2

Step 1: List all unique values in order.

> Looking at the data, the smallest value is 0 and the largest is 5. So the unique values are: 0, 1, 2, 3, 4, 5.

Step 2: Count how many times each value appears (tally).

> Value 0: appears at positions 3, 9, 15 → count = 3

> Value 1: appears at positions 2, 5, 8, 11, 14, 18 → count = 6

> Value 2: appears at positions 1, 6, 10, 13, 16, 20 → count = 6

Value 3: appears at positions 4, 12, 19 → count = 3

Value 4: appears at position 7 → count = 1

Value 5: appears at position 17 → count = 1

Step 3: Check that the frequencies add up to n.

3 + 6 + 6 + 3 + 1 + 1 = 20 □ (matches our 20 students)

Step 4: Calculate relative frequencies.

For 0: 3 ÷ 20 = 0.15

For 1: 6 ÷ 20 = 0.30

For 2: 6 ÷ 20 = 0.30

For 3: 3 ÷ 20 = 0.15

For 4: 1 ÷ 20 = 0.05

For 5: 1 ÷ 20 = 0.05

Check: 0.15 + 0.30 + 0.30 + 0.15 + 0.05 + 0.05 = 1.00 □

Step 5: Calculate cumulative frequencies.

For 0: 3

For 1: 3 + 6 = 9

For 2: 9 + 6 = 15

For 3: 15 + 3 = 18

For 4: 18 + 1 = 19

For 5: 19 + 1 = 20 (must equal n) □

x (siblings)TallyfRelative fCumulative f 0|||30.153 1|||| |60.309 2|||| |60.3015 3|||30.1518 4|10.0519 5|10.0520 **Total201.00**

4.3 Grouped Frequency Tables (Continuous Data)

When data values are spread over a wide range (like exam scores from 12 to 95), listing every individual value would create a table with too many rows to be useful. Instead, we group the data into **class intervals** (also called bins or classes).

Key Terminology

- **Class interval:** A range of values, e.g. 10–19, 20–29.

- **Lower class limit:** The smallest value that can belong to the class (e.g. 10 in "10–19").
- **Upper class limit:** The largest value that can belong to the class (e.g. 19 in "10–19").
- **Class boundaries:** The true edges of the class, found by going halfway between the upper limit of one class and the lower limit of the next. For "10–19" and "20–29", the boundary between them is 19.5. So the boundaries for "10–19" are 9.5 and 19.5.
- **Class width (w):** The difference between consecutive lower class limits (or upper class limits). For 10–19, 20–29: width = 20 − 10 = 10.
- **Class midpoint (x):** The centre of the class, calculated as (lower limit + upper limit) / 2. For 10–19: midpoint = (10 + 19) ÷ 2 = 14.5.

How Many Classes?
A common rule is Sturges' Rule: $k = 1 + 3.322 \log_{10}(n)$, where n is the number of data values and k is the number of classes (round up). In practice, STA1610 exams usually give you the number of classes or the class width, so you will not need to calculate this from scratch. Use between 5 and 15 classes as a general guideline.

[CONCEPT] Constructing a grouped frequency table

The following are the marks (out of 100) obtained by 30 students in a statistics test:
12, 18, 22, 25, 28, 31, 33, 35, 37, 40, 42, 44, 45, 47, 48, 50, 52, 54, 55, 58, 60, 62, 65, 67, 70, 72, 75, 78, 82, 91
Construct a grouped frequency table using the classes 10–19, 20–29, 30–39, 40–49, 50–59, 60–69, 70–79, 80–89, 90–99.

Step 1: Determine the class width and midpoints.

Class width = 20 − 10 = 10 (all classes have the same width).

Midpoint of 10–19: (10 + 19) ÷ 2 = 29 ÷ 2 = 14.5

Midpoint of 20–29: (20 + 29) ÷ 2 = 49 ÷ 2 = 24.5

Midpoint of 30–39: (30 + 39) ÷ 2 = 69 ÷ 2 = 34.5

Midpoint of 40–49: (40 + 49) ÷ 2 = 89 ÷ 2 = 44.5

Midpoint of 50–59: (50 + 59) ÷ 2 = 109 ÷ 2 = 54.5

Midpoint of 60–69: (60 + 69) ÷ 2 = 129 ÷ 2 = 64.5

Midpoint of 70–79: (70 + 79) ÷ 2 = 149 ÷ 2 = 74.5

Midpoint of 80–89: (80 + 89) ÷ 2 = 169 ÷ 2 = 84.5

Midpoint of 90–99: (90 + 99) ÷ 2 = 189 ÷ 2 = 94.5

Step 2: Sort each data value into its class and count.

10–19: 12, 18 → f = 2

20–29: 22, 25, 28 → f = 3

30–39: 31, 33, 35, 37 → f = 4

40–49: 40, 42, 44, 45, 47, 48 → f = 6

50–59: 50, 52, 54, 55, 58 → f = 5

60–69: 60, 62, 65, 67 → f = 4

70–79: 70, 72, 75, 78 → f = 4

80–89: 82 → f = 1

90–99: 91 → f = 1

Step 3: Verify the total.

2 + 3 + 4 + 6 + 5 + 4 + 4 + 1 + 1 = 30 □

ClassMidpoint (x)fCumulative fRelative f 10–1914.5220.067 20–2924.5350.100 30–3934.5490.133 40–4944.56150.200 50–5954.55200.167 60–6964.54240.133 70–7974.54280.133 80–8984.51290.033 90–9994.51300.033 **Total30≈1.00**

Common Exam Mistake: Students often confuse class limits with class boundaries. If a question asks for class boundaries of "10–19, 20–29", the boundaries are 9.5, 19.5, 29.5, etc. — NOT 10, 19, 20, 29. The boundaries sit halfway between adjacent limits.

4.4 Reading Information from a Frequency Table

STA1610 exams frequently ask you to extract specific information from a frequency table. Here are the most common question types:

- **"How many students scored less than 50?"** → Look at the cumulative frequency for the class ending at 49. From our table above: cumulative f for 40–49 = 15. So 15 students scored less than 50.
- **"What percentage of students scored 60 or more?"** → Total students = 30. Students scoring 60+ = 4 + 4 + 1 + 1 = 10. Percentage = (10 ÷ 30) × 100 = 33.3%.
- **"Which class has the highest frequency?"** → The class 40–49 with f = 6. This is called the **modal class**.

Exam Tip: STA1610 papers almost always include at least one question requiring you to complete a partially filled frequency table (some frequencies missing) and then use it to calculate a mean or median. Practise filling in blanks using the constraint that all frequencies must sum to n.

TOPIC 05

Measures of Central Tendency

Finding the "centre" of a dataset — Mean, Median, Mode

5.1 What Does "Central Tendency" Mean?

When you have a set of numbers, you often want to summarise the entire set with a single number that represents the "typical" or "middle" value. This is what measures of central tendency do. There are three main measures, and each one defines "centre" in a slightly different way:

- **Mean:** The arithmetic average. Add everything up and divide by how many there are.
- **Median:** The middle value when the data is sorted from smallest to largest. Half the data is above it, half is below.
- **Mode:** The value that appears most frequently.

> Analogy: Imagine five friends comparing their monthly phone bills: R80, R90, R100, R110, and R5 000 (one friend went on international roaming). The mean would be pulled up to about R1 076 by that one huge bill, which does not represent anyone. The median would be R100, which is much more typical. The mode would not exist here since no value repeats. This shows why we need all three measures — they each tell a different story.

5.2 The Mean

5.2.1 Mean from Raw Data

The mean is the sum of all values divided by the number of values. We use different symbols depending on whether we are working with a population or a sample.

> Population Mean
> $\mu = (\Sigma_{i=1}^{N} x_i) / N$ where μ (Greek letter "mu") = population mean, x_i = each individual value, and N = total number of values in the population.
>
> Sample Mean
> $\bar{x} = (\Sigma_{i=1}^{n} x_i) / n$ where $\bar{x}$ ("x-bar") = sample mean, and n = number of values in the sample.

In STA1610, you will almost always work with sample data, so you will use $\bar{x}$. The calculation is identical; only the symbol changes.

[CONCEPT] Calculating the mean from raw data

Find the mean of the following test scores: 14, 22, 18, 25, 31, 20, 17.

Step 1: Identify n (the number of values).

> Counting the values: 14, 22, 18, 25, 31, 20, 17 → n = 7

Step 2: Calculate the sum of all values (Σ x).

> 14 + 22 = 36

> 36 + 18 = 54

54 + 25 = 79

79 + 31 = 110

110 + 20 = 130

130 + 17 = 147

So Σ x = 147

Step 3: Divide the sum by n.

$\bar{x} = 147 / 7$

147 ÷ 7 = 21

$\bar{x} = 21$

5.2.2 Mean from an Ungrouped Frequency Table

When data is presented in a frequency table, you do not add up all the individual values one by one. Instead, you multiply each unique value by its frequency, sum those products, and divide by the total frequency.

Mean from Frequency Table
$\bar{x} = (\Sigma f \cdot x) / \Sigma f = \Sigma fx / n$ where x = each unique value, f = its frequency, and $n = \Sigma f$.

[CONCEPT] Mean from an ungrouped frequency table

Using the siblings data from Topic 4, calculate the mean number of siblings.

Step 1: Set up the fx column.

x = 0, f = 3 → fx = 0 × 3 = 0

x = 1, f = 6 → fx = 1 × 6 = 6

x = 2, f = 6 → fx = 2 × 6 = 12

x = 3, f = 3 → fx = 3 × 3 = 9

x = 4, f = 1 → fx = 4 × 1 = 4

x = 5, f = 1 → fx = 5 × 1 = 5

Step 2: Sum the fx column.

$\Sigma fx = 0 + 6 + 12 + 9 + 4 + 5 = 36$

Step 3: Divide by n.

$n = \Sigma f = 20$

$\bar{x} = 36 / 20$

$36 \div 20 = 1.8$

$\bar{x} = 1.8$ siblings per student

5.2.3 Mean from a Grouped Frequency Table

With grouped data, we no longer know the exact values — we only know which class each value falls into. So we use the **class midpoint** as a representative value for every observation in that class.

Mean from Grouped Data
$\bar{x} \approx (\Sigma f \cdot x_m) / \Sigma f$ where x_m = class midpoint and f = class frequency. The result is an approximation because we assume all values in a class equal the midpoint.

[EXAM] Mean from grouped data (STA1610 style)

Using the grouped frequency table from Topic 4 (30 students' marks), estimate the mean mark.

Step 1: Multiply each midpoint by its frequency.

$14.5 \times 2 = 29.0$

$24.5 \times 3 = 73.5$

$34.5 \times 4 = 138.0$

$44.5 \times 6 = 267.0$

$54.5 \times 5 = 272.5$

$64.5 \times 4 = 258.0$

$74.5 \times 4 = 298.0$

$84.5 \times 1 = 84.5$

$94.5 \times 1 = 94.5$

Step 2: Sum the fx column.

$29.0 + 73.5 = 102.5$

$102.5 + 138.0 = 240.5$

$240.5 + 267.0 = 507.5$

$507.5 + 272.5 = 780.0$

$780.0 + 258.0 = 1\ 038.0$

$1\ 038.0 + 298.0 = 1\ 336.0$

1 336.0 + 84.5 = 1 420.5

1 420.5 + 94.5 = 1 515.0

Σ fx = 1 515.0

Step 3: Divide by n.

$\bar{x}$ = 1 515 / 30

1 515 ÷ 30 = 50.5

$\bar{x} \approx 50.5$ marks

5.3 The Median

The median is the value that sits exactly in the middle when all data is arranged in ascending order. It splits the dataset into two equal halves: 50% of values are below the median and 50% are above.

5.3.1 Median from Raw Data

Finding the Median Position
Position of the median = (n + 1) / 2

If n is odd, this gives a whole number and the median is the value at that position.
If n is even, this gives a decimal (e.g. 5.5), meaning the median is the average of the two values surrounding that position.

[CONCEPT] Median with odd n

Find the median of: 14, 22, 18, 25, 31, 20, 17.

Step 1: Sort the data in ascending order.

14, 17, 18, 20, 22, 25, 31

Step 2: Find the median position.

n = 7

Position = (7 + 1) ÷ 2 = 8 ÷ 2 = 4th position

Step 3: Read the value at the 4th position.

1st = 14, 2nd = 17, 3rd = 18, **4th = 20**

Median = 20

[CONCEPT] Median with even n

Find the median of: 5, 8, 12, 15, 20, 25.

Step 1: Data is already sorted.

5, 8, 12, 15, 20, 25

Step 2: Find the median position.

n = 6

Position = (6 + 1) ÷ 2 = 7 ÷ 2 = 3.5th position

This means the median falls between the 3rd and 4th values.

Step 3: Average the 3rd and 4th values.

3rd value = 12, 4th value = 15

Median = (12 + 15) ÷ 2 = 27 ÷ 2 = 13.5

Median = 13.5

5.3.2 Median from Grouped Data (Interpolation)

This is the version STA1610 exams test most often. Since we do not know the exact values inside each class, we estimate the median using a formula that assumes the data is evenly spread within the median class.

Median from Grouped Data
Median = $L + ((n / 2 - F) / (f_m)) \times w$ where:
L = lower class boundary of the median class
n = total frequency
F = cumulative frequency of the class before the median class
f_m = frequency of the median class
w = class width

How to find the median class: Calculate n/2. The median class is the first class whose cumulative frequency is greater than or equal to n/2. Think of it this way: you are looking for the class that contains the "middle" observation.

[EXAM] Median from grouped data (full interpolation)

Using the grouped frequency table from Topic 4 (30 students' marks), estimate the median mark.

Step 1: Find n/2 to locate the median class.

n = 30

n/2 = 30 ÷ 2 = 15

We need the first class whose cumulative frequency ≥ 15.

Step 2: Scan the cumulative frequencies.

10–19: cum f = 2 (not yet 15)

20–29: cum f = 5 (not yet)

30–39: cum f = 9 (not yet)

40–49: cum f = 15 □ (this is ≥ 15, so this is the median class)

Step 3: Identify the values for the formula.

Median class = 40–49

L = lower class boundary = 39.5 (halfway between 39 and 40)

F = cumulative frequency before median class = 9 (cum f of 30–39)

f_m = frequency of median class = 6

w = class width = 10

Step 4: Substitute into the formula.

Median = 39.5 + ((15 - 9) / 6) × 10

= 39.5 + (6 / 6) × 10

= 39.5 + (1) × 10

= 39.5 + 10

= 49.5

Median ≈ 49.5 marks

Common Exam Mistake: Students frequently use the lower class limit (40) instead of the lower class boundary (39.5) in the median formula. This gives the wrong answer. Always use boundaries: subtract 0.5 from the lower limit when classes are whole numbers.

5.4 The Mode

The mode is the simplest measure: it is the value that occurs most often. A dataset can have one mode (unimodal), two modes (bimodal), or no mode at all (if no value repeats).

5.4.1 Mode from Raw Data or Ungrouped Frequency Table

Simply look for the value with the highest frequency. From our siblings data, both 1 and 2 appear 6 times, so the data is bimodal with modes 1 and 2.

5.4.2 Modal Class from Grouped Data

For grouped data, we identify the **modal class** — the class with the highest frequency. From our marks table, the modal class is 40–49 (f = 6).

Mode from Grouped Data (Estimation)
Mode ≈ L + ((d$_1$) / (d$_1$ + d$_2$)) × w where:
L = lower class boundary of the modal class
d$_1$ = frequency of modal class minus frequency of the class before it
d$_2$ = frequency of modal class minus frequency of the class after it
w = class width

[EXAM] Mode from grouped data

Estimate the mode for the marks data.

Step 1: Identify the modal class.

Highest frequency = 6, in class 40–49.

Step 2: Identify the values.

L = 39.5 (lower boundary of 40–49)

Frequency of modal class = 6

Frequency of class before (30–39) = 4 → $d_1 = 6 - 4 = 2$

Frequency of class after (50–59) = 5 → $d_2 = 6 - 5 = 1$

w = 10

Step 3: Substitute.

Mode = 39.5 + (2 / (2 + 1)) × 10

= 39.5 + (2 / 3) × 10

= 39.5 + 0.6667 × 10

= 39.5 + 6.667

= 46.167

Mode ≈ 46.2 marks

5.5 Comparing Mean, Median, and Mode (Skewness)

The relationship between the three measures tells you about the shape of the data distribution:

Distribution Shape	Relationship	What It Means
Symmetrical	Mean ≈ Median ≈ Mode	Data is balanced on both sides
Right-skewed (positive)	Mode < Median < Mean	A tail of high values pulls the mean up
Left-skewed (negative)	Mean < Median < Mode	A tail of low values pulls the mean down

Exam Tip: STA1610 frequently asks "Which measure of central tendency is most appropriate for skewed data?" The answer is the median, because it is not affected by extreme values (outliers). The mean gets pulled toward the tail, making it misleading.

TOPIC 06

Measures of Dispersion

How spread out is the data? Range, Variance, Standard Deviation

6.1 Why Do We Need Measures of Dispersion?

Central tendency tells you where the centre is, but it tells you nothing about how spread out the data is around that centre. Consider two classes that both have a mean exam score of 60:

- **Class A:** 58, 59, 60, 61, 62 (everyone scored close to 60)
- **Class B:** 20, 40, 60, 80, 100 (scores are wildly spread out)

Both have the same mean, but the experience in each class is completely different. Measures of dispersion quantify this spread.

6.2 The Range

The range is the simplest measure of spread. It is the difference between the largest and smallest values.

Range = x_{max} - x_{min}

[CONCEPT] Calculating the range

Find the range of: 14, 22, 18, 25, 31, 20, 17.

Step 1: Identify the maximum and minimum.

Maximum = 31, Minimum = 14

Step 2: Subtract.

Range = 31 − 14 = 17

Range = 17

Limitation: The range uses only two values and ignores everything else. A single outlier can make the range enormous even if the rest of the data is tightly clustered. That is why we need better measures.

6.3 Variance and Standard Deviation

These are the most important measures of dispersion in statistics. They measure, on average, how far each data point is from the mean.

6.3.1 The Logic Behind Variance

You might think we could just calculate the average distance from the mean: add up all the $(x_i - \bar{x})$ values and divide by n. But there is a problem — the positive and negative deviations always cancel out to exactly zero. That is a mathematical property of the mean.

To fix this, we **square** each deviation before averaging. Squaring makes everything positive and also gives more weight to values that are far from the mean (which is desirable — outliers should have a bigger impact on our measure of spread).

Population Variance
$\sigma^2 = (\Sigma_{i=1}^{N}(x_i - \mu)^2) / N$

Sample Variance
$s^2 = (\Sigma_{i=1}^{n}(x_i - \bar{x})^2) / (n - 1)$

Standard Deviation = the square root of the variance:
$\sigma = \sqrt{(\sigma^2)}$ or $s = \sqrt{(s^2)}$

Why do we divide by $(n - 1)$ for a sample? When you use a sample to estimate the population variance, dividing by n tends to underestimate the true variance. This is because the sample mean $\bar{x}$ is itself calculated from the same data, so the deviations from $\bar{x}$ are slightly smaller than the deviations from the true population mean μ. Dividing by $(n - 1)$ instead of n corrects for this underestimation. The quantity $(n - 1)$ is called the degrees of freedom. In STA1610, you will almost always use the sample formula with $(n - 1)$.

6.3.2 Step-by-Step Calculation from Raw Data

[EXAM] Variance and standard deviation from raw data

Calculate the sample variance and standard deviation of: 4, 8, 6, 5, 7.

Step 1: Calculate the sample mean.

$\Sigma x = 4 + 8 + 6 + 5 + 7 = 30$

$n = 5$

$\bar{x} = 30 \div 5 = 6$

Step 2: Calculate each deviation from the mean $(x_i - \bar{x})$.

$4 - 6 = -2$

$8 - 6 = 2$

$6 - 6 = 0$

$5 - 6 = -1$

$7 - 6 = 1$

Step 3: Square each deviation $(x_i - \bar{x})^2$.

$(-2)^2 = 4$

$(2)^2 = 4$

$(0)^2 = 0$

$(-1)^2 = 1$

$(1)^2 = 1$

Step 4: Sum the squared deviations.

$\Sigma(x_i - \bar{x})^2 = 4 + 4 + 0 + 1 + 1 = 10$

Step 5: Divide by $(n - 1)$ to get the sample variance.

$n - 1 = 5 - 1 = 4$

$s^2 = 10 / 4 = 2.5$

Step 6: Take the square root to get the standard deviation.

$s = \sqrt{(2.5)}$

$s = 1.5811$ (to 4 decimal places)

$s^2 = 2.5$, $s \approx 1.58$

6.3.3 Computational (Shortcut) Formula

The definitional formula above is great for understanding, but it is tedious with large datasets or messy means. STA1610 exams often expect you to use the computational formula, which avoids calculating individual deviations:

Computational Formula for Sample Variance
$s^2 = (\Sigma x^2 - ((\Sigma x)^2) / n) / (n - 1)$ This gives exactly the same answer as the definitional formula but is faster to compute, especially on a calculator.

[EXAM] Variance using the computational formula

Verify the variance of 4, 8, 6, 5, 7 using the shortcut formula.

Step 1: Calculate Σx and Σx^2.

$\Sigma x = 4 + 8 + 6 + 5 + 7 = 30$

$\Sigma x^2 = 4^2 + 8^2 + 6^2 + 5^2 + 7^2$

$= 16 + 64 + 36 + 25 + 49$

$= 190$

Step 2: Substitute into the formula.

$s^2 = (190 - (30^2) / 5) / (5 - 1)$

$= (190 - 900 / 5) / 4$

$= (190 - 180) / 4$

$= 10 / 4$

$= 2.5$ □ (matches our previous answer)

$s^2 = 2.5$

6.4 Variance and Standard Deviation from a Frequency Table

From an Ungrouped Frequency Table
$s^2 = (\Sigma f x^2 - ((\Sigma fx)^2) / n) / (n - 1)$ where fx^2 means $f \times x^2$ (square x first, then multiply by f).

[EXAM] Standard deviation from a frequency table

The number of defects found in 40 manufactured items is summarised below. Calculate the sample standard deviation.

x (defects)f 010 114 29 35 42

Step 1: Build the fx and fx^2 columns.

x=0: fx = 0×10 = 0, $fx^2 = 0^2 \times 10 = 0$

x=1: fx = 1×14 = 14, $fx^2 = 1^2 \times 14 = 14$

x=2: fx = 2×9 = 18, $fx^2 = 2^2 \times 9 = 4 \times 9 = 36$

x=3: fx = 3×5 = 15, $fx^2 = 3^2 \times 5 = 9 \times 5 = 45$

x=4: fx = 4×2 = 8, $fx^2 = 4^2 \times 2 = 16 \times 2 = 32$

Step 2: Sum the columns.

$n = \Sigma f = 10 + 14 + 9 + 5 + 2 = 40$

$\Sigma fx = 0 + 14 + 18 + 15 + 8 = 55$

$\Sigma fx^2 = 0 + 14 + 36 + 45 + 32 = 127$

Step 3: Apply the formula.

$s^2 = (127 - (55^2) / 40) / (40 - 1)$

$= (127 - 3025 / 40) / 39$

$= (127 - 75.625) / 39$

$= 51.375 / 39$

$= 1.3173$

Step 4: Take the square root.

$s = \sqrt{(1.3173)} = 1.1477$

$s^2 \approx 1.32$, $s \approx 1.15$ defects

6.5 Variance from Grouped Data

For grouped data, we use the class midpoints as representative values, exactly as we did for the grouped mean.

$s^2 \approx (\Sigma f x_m^2 - ((\Sigma f x_m)^2) / n) / (n - 1)$ where x_m = class midpoint.

[EXAM] Standard deviation from grouped data

Estimate the standard deviation for the 30 students' marks from Topic 4.

Step 1: We already know $\Sigma fx_m = 1\ 515$ and $n = 30$ from the mean calculation.

Step 2: Calculate $\Sigma f x_m^2$.

$14.5^2 \times 2 = 210.25 \times 2 = 420.50$

$24.5^2 \times 3 = 600.25 \times 3 = 1\,800.75$

$34.5^2 \times 4 = 1\,190.25 \times 4 = 4\,761.00$

$44.5^2 \times 6 = 1\,980.25 \times 6 = 11\,881.50$

$54.5^2 \times 5 = 2\,970.25 \times 5 = 14\,851.25$

$64.5^2 \times 4 = 4\,160.25 \times 4 = 16\,641.00$

$74.5^2 \times 4 = 5\,550.25 \times 4 = 22\,201.00$

$84.5^2 \times 1 = 7\,140.25 \times 1 = 7\,140.25$

$94.5^2 \times 1 = 8\,930.25 \times 1 = 8\,930.25$

Step 3: Sum the fx^2 column.

$420.50 + 1\,800.75 = 2\,221.25$

$2\,221.25 + 4\,761.00 = 6\,982.25$

$6\,982.25 + 11\,881.50 = 18\,863.75$

$18\,863.75 + 14\,851.25 = 33\,715.00$

$33\,715.00 + 16\,641.00 = 50\,356.00$

$50\,356.00 + 22\,201.00 = 72\,557.00$

$72\,557.00 + 7\,140.25 = 79\,697.25$

$79\,697.25 + 8\,930.25 = 88\,627.50$

$\Sigma fx_m^{\,2} = 88\,627.50$

Step 4: Apply the formula.

$s^2 = (88\,627.50 - (1\,515^2) / 30) / (30 - 1)$

$= (88\,627.50 - 2\,295\,225 / 30) / 29$

$= (88\,627.50 - 76\,507.50) / 29$

$= 12\,120.00 / 29$

$= 417.931$

Step 5: Standard deviation.

$s = \sqrt{(417.931)} = 20.443$

$s^2 \approx 417.93$, $s \approx 20.44$ marks

6.6 The Coefficient of Variation (CV)

The standard deviation is measured in the same units as the data (marks, rands, kilograms, etc.). This makes it hard to compare the spread of two datasets that have very different means or different units. The coefficient of variation solves this by expressing the standard deviation as a percentage of the mean.

CV = s / $\bar{x}$ × 100% A higher CV means greater relative variability.

[EXAM] Comparing variability using CV

Class A has $\bar{x}$ = 60, s = 5. Class B has $\bar{x}$ = 80, s = 8. Which class has more relative variability?

Step 1: Calculate CV for each class.

CV_A = (5 ÷ 60) × 100 = 8.33%

CV_B = (8 ÷ 80) × 100 = 10.00%

Step 2: Compare.

10.00% > 8.33%, so Class B has more relative variability even though its standard deviation (8) is not dramatically larger than Class A's (5).

Class B is more variable relative to its mean.

Exam Tip: STA1610 exams love asking you to compare two datasets using the CV. The key phrase in the question is usually "which dataset is more consistent?" — the one with the lower CV is more consistent (less variable). Do not just compare standard deviations; always compute the CV when the means differ.

6.7 Summary of Dispersion Measures

Measure	Formula (Sample)	Units	
Range	$x_{max} - x_{min}$	Quick overview; sensitive to outliers	Same as data
Variance	$s^2 = (\Sigma(x-\bar{x})^2) / (n-1)$	Mathematical work; foundation for other tests	Squared units
Std Deviation	$s = \sqrt{(s^2)}$	Most common measure of spread	Same as data
Coeff of Variation	s / $\bar{x}$ × 100%	Comparing spread across different datasets	Percentage

Common Exam Mistake: Confusing variance with standard deviation. If the question asks for the standard deviation and you stop at the variance (or vice versa), you lose marks. Always check what is being asked. Remember: variance = s^2 (squared units), standard deviation = s (original units). The standard deviation is almost always the more interpretable number.

TOPIC 07

Measures of Position & Box Plots

Quartiles, percentiles, and visualising the five-number summary

7.1 What Are Measures of Position?

Measures of central tendency tell you where the middle is. Measures of dispersion tell you how spread out the data is. Measures of position tell you *where a specific value sits relative to the rest of the data*. For example, if you scored 72% on an exam, is that good? It depends on where 72% falls compared to everyone else. If 90% of students scored below you, your position is excellent. If only 20% scored below you, your position is poor — even though 72% sounds like a decent mark.

The most common measures of position are **quartiles**, **percentiles**, and **deciles**.

7.2 Quartiles

Quartiles divide a sorted dataset into four equal parts, each containing 25% of the data.

- **Q_1 (First Quartile / Lower Quartile):** 25% of the data falls below this value.
- **Q_2 (Second Quartile):** This is exactly the **median**. 50% of the data falls below it.
- **Q_3 (Third Quartile / Upper Quartile):** 75% of the data falls below this value.

Quartile Positions (for raw data)
Position of Q_1 = (n + 1) / 4
Position of Q_2 = (2(n + 1)) / 4 = (n + 1) / 2 (same as the median)
Position of Q_3 = (3(n + 1)) / 4

If the position is a whole number, the quartile is the value at that position. If the position is a decimal, interpolate between the two surrounding values.

[CONCEPT] Calculating quartiles from raw data

Find Q_1, Q_2, and Q_3 for the following 11 exam scores:
45, 52, 58, 63, 67, 71, 74, 78, 82, 88, 95

Step 1: Verify the data is sorted.

45, 52, 58, 63, 67, 71, 74, 78, 82, 88, 95 → already in ascending order □

n = 11

Step 2: Find Q_1.

Position of Q_1 = (11 + 1) ÷ 4 = 12 ÷ 4 = 3rd position

The 3rd value in the sorted list is 58.

$Q_1 = 58$

Step 3: Find Q_2 (the median).

Position of $Q_2 = (11 + 1) \div 2 = 12 \div 2 = 6$th position

The 6th value is 71.

$Q_2 = 71$

Step 4: Find Q_3.

Position of $Q_3 = 3(11 + 1) \div 4 = 36 \div 4 = 9$th position

The 9th value is 82.

$Q_3 = 82$

$Q_1 = 58, Q_2 = 71, Q_3 = 82$

[CONCEPT] Quartiles with interpolation (even n)

Find Q_1 and Q_3 for: 10, 15, 20, 25, 30, 35, 40, 45, 50, 55 ($n = 10$).

Step 1: Q_1 position.

$(10 + 1) \div 4 = 11 \div 4 = 2.75$th position

This falls between the 2nd value (15) and the 3rd value (20).

Interpolate: $Q_1 = 15 + 0.75 \times (20 - 15)$

$= 15 + 0.75 \times 5$

$= 15 + 3.75$

$= 18.75$

Step 2: Q_3 position.

$3(10 + 1) \div 4 = 33 \div 4 = 8.25$th position

This falls between the 8th value (45) and the 9th value (50).

Interpolate: $Q_3 = 45 + 0.25 \times (50 - 45)$

$= 45 + 0.25 \times 5$

$= 45 + 1.25$

$= 46.25$

$Q_1 = 18.75$, $Q_3 = 46.25$

7.3 The Interquartile Range (IQR)

The IQR measures the spread of the middle 50% of the data. Unlike the range, it is not affected by extreme outliers because it ignores the bottom 25% and the top 25%.

$$IQR = Q_3 - Q_1$$

[CONCEPT] IQR calculation

Using the quartiles from the first example above ($Q_1 = 58$, $Q_3 = 82$), find the IQR.

Step 1: Subtract.

$$IQR = 82 - 58 = 24$$

IQR = 24 marks. The middle 50% of students scored within a 24-mark range.

7.4 Percentiles

Percentiles divide the data into 100 equal parts. The k-th percentile (P_k) is the value below which k% of the data falls. Quartiles are just special percentiles: $Q_1 = P_{25}$, $Q_2 = P_{50}$, $Q_3 = P_{75}$.

Position of the k-th Percentile

$$\text{Position} = k / 100 \times (n + 1)$$

[EXAM] Finding a percentile

Find the 90th percentile (P_{90}) of the 11 exam scores: 45, 52, 58, 63, 67, 71, 74, 78, 82, 88, 95.

Step 1: Calculate the position.

$$\text{Position} = (90 \div 100) \times (11 + 1) = 0.9 \times 12 = 10.8\text{th position}$$

Step 2: Interpolate between the 10th and 11th values.

10th value = 88, 11th value = 95

$$P_{90} = 88 + 0.8 \times (95 - 88)$$

$$= 88 + 0.8 \times 7$$

$$= 88 + 5.6$$

$$= 93.6$$

$P_{90} = 93.6$. This means 90% of students scored below 93.6 marks.

7.5 Quartiles from Grouped Data

Just as we interpolated to find the median from grouped data, we can find Q_1 and Q_3 using a similar formula.

Quartiles from Grouped Data
$Q_k = L + ((kn / 4 - F) / (f_q)) \times w$ where k = 1 for Q_1, k = 3 for Q_3, and the other symbols are the same as in the median formula (L = lower boundary of the quartile class, F = cumulative frequency before that class, f_q = frequency of the quartile class, w = class width).

[EXAM] Q_1 from grouped data

Using the marks data from Topic 4 (n = 30), estimate Q_1.

Step 1: Find the Q_1 position.

$kn/4 = 1 \times 30 \div 4 = 7.5$

We need the first class whose cumulative frequency ≥ 7.5.

Step 2: Scan cumulative frequencies.

10–19: cum f = 2 (not yet)

20–29: cum f = 5 (not yet)

30–39: cum f = 9 ≥ 7.5 □ → Q_1 class is 30–39

Step 3: Identify formula values.

L = 29.5 (lower boundary of 30–39)

F = 5 (cum f before Q_1 class, i.e. cum f of 20–29)

f_q = 4 (frequency of 30–39)

w = 10

Step 4: Substitute.

$Q_1 = 29.5 + ((7.5 - 5) / 4) \times 10$

$= 29.5 + (2.5 / 4) \times 10$

$= 29.5 + 0.625 \times 10$

$= 29.5 + 6.25$

$= 35.75$

$Q_1 \approx 35.75$ marks

7.6 The Five-Number Summary & Box Plots

The five-number summary condenses an entire dataset into five key values:

Five-Number Summary
x_{min}, Q_1, Q_2 (Median), Q_3, x_{max}

A **box plot** (also called a box-and-whisker plot) is a visual representation of the five-number summary. It consists of:

- A **box** stretching from Q_1 to Q_3 (representing the middle 50% of the data).
- A **line inside the box** at the median (Q_2).
- **Whiskers** extending from the box to the minimum and maximum values (or to the nearest non-outlier values, depending on the convention).

[EXAM] Five-number summary and outlier detection

For the 11 exam scores (45, 52, 58, 63, 67, 71, 74, 78, 82, 88, 95), state the five-number summary and check for outliers.

Step 1: State the five-number summary.

Min = 45

$Q_1 = 58$ (calculated earlier)

Median = 71

$Q_3 = 82$

Max = 95

Step 2: Calculate the IQR.

$IQR = Q_3 - Q_1 = 82 - 58 = 24$

Step 3: Find the outlier fences.

Lower fence $= Q_1 - 1.5 \times IQR = 58 - 1.5 \times 24 = 58 - 36 = 22$

Upper fence $= Q_3 + 1.5 \times IQR = 82 + 1.5 \times 24 = 82 + 36 = 118$

Step 4: Check for outliers.

Any value below 22 or above 118 is an outlier.

Min = 45 (above 22, not an outlier)

Max = 95 (below 118, not an outlier)

No outliers in this dataset.

Five-number summary: 45, 58, 71, 82, 95. No outliers.

Exam Tip: STA1610 exams frequently ask you to (a) calculate the five-number summary, (b) draw or interpret a box plot, and (c) identify outliers using the 1.5 × IQR rule. Make sure you can do all three. When asked to "comment on the shape" from a box plot: if the median line is centred in the box and the whiskers are equal length, the data is roughly symmetrical. If the median is closer to Q_1 and the right whisker is longer, the data is right-skewed.

Common Exam Mistake: Students often confuse the outlier fences with the whisker endpoints. The fences are the boundaries for identifying outliers ($Q_1 - 1.5\times IQR$ and $Q_3 + 1.5\times IQR$). The whiskers on the box plot extend to the most extreme data points that are not outliers — they do not necessarily reach the fences themselves.

TOPIC 08

Probability Basics

The language of chance — experiments, sample spaces, and events

8.1 What Is Probability?

Probability is a number between 0 and 1 that measures how likely an event is to occur. A probability of 0 means the event is impossible. A probability of 1 means the event is certain. A probability of 0.5 means the event is equally likely to happen or not happen.

Everyday analogy: When you check the weather app and it says "70% chance of rain," it is giving you a probability of 0.70. It does not mean it will rain for 70% of the day. It means that, based on the available evidence, the meteorologists believe rain is quite likely but not guaranteed. Probability quantifies uncertainty.

8.2 Key Terminology

Before we can calculate probabilities, we need to understand the vocabulary. Every probability problem starts with these building blocks:

- **Random Experiment:** A process that produces an outcome that cannot be predicted with certainty in advance. Examples: tossing a coin, rolling a die, drawing a card from a deck.
- **Sample Space (S):** The set of all possible outcomes of the experiment. For a coin toss: S = {H, T}. For a six-sided die: S = {1, 2, 3, 4, 5, 6}.
- **Event:** A subset of the sample space — one or more outcomes that we are interested in. For example, "rolling an even number" is the event E = {2, 4, 6}.
- **Simple Event:** An event consisting of exactly one outcome (e.g. rolling a 3).
- **Compound Event:** An event consisting of two or more outcomes (e.g. rolling a number greater than 4 = {5, 6}).

8.3 Three Approaches to Probability

8.3.1 Classical (Theoretical) Probability

Used when all outcomes are equally likely. This is the approach you will use most in STA1610.

P(E) = Number of favourable outcomes / Total number of possible outcomes = n(E) / n(S)

[CONCEPT] Classical probability with a die

A fair six-sided die is rolled. Find the probability of (a) rolling a 4, (b) rolling an odd number, (c) rolling a number less than 7.

Step 1: State the sample space.

> S = {1, 2, 3, 4, 5, 6}, so n(S) = 6

Step 2: Part (a) — rolling a 4.

> Event A = {4}, so n(A) = 1

> P(A) = 1 ÷ 6 = 1/6 ≈ 0.1667

Step 3: Part (b) — rolling an odd number.

> Event B = {1, 3, 5}, so n(B) = 3

> P(B) = 3 ÷ 6 = 1/2 = 0.5

Step 4: Part (c) — rolling a number less than 7.

> Event C = {1, 2, 3, 4, 5, 6}, so n(C) = 6

> P(C) = 6 ÷ 6 = 1 (this is a certain event — every outcome is less than 7)

> P(4) = 1/6, P(odd) = 1/2, P(<7) = 1

8.3.2 Relative Frequency (Empirical) Probability

Used when outcomes are not equally likely or when we rely on observed data. The probability is estimated from the proportion of times the event occurred in past trials.

> P(E) ≈ Number of times E occurred / Total number of trials

[CONCEPT] Empirical probability

A factory inspected 500 items and found 15 were defective. Estimate the probability that a randomly selected item is defective.

Step 1: Identify the counts.

> Defective items = 15, Total items = 500

Step 2: Calculate.

> P(defective) = 15 ÷ 500 = 0.03

> P(defective) = 0.03 (or 3%)

8.3.3 Subjective Probability

Based on personal judgement, experience, or intuition. For example, "I think there is an 80% chance my team will win on Saturday." This is not calculated from data or theory. STA1610 rarely tests this, but you should know it exists.

8.4 Fundamental Properties of Probability

Three Axioms of Probability
1. $0 \leq P(E) \leq 1$ for any event E. (Probabilities are always between 0 and 1.)
2. $P(S) = 1$. (The probability of the entire sample space is 1 — something must happen.)
3. If A and B are mutually exclusive events, then P(A or B) = P(A) + P(B).

What does "mutually exclusive" mean? Two events are mutually exclusive (also called disjoint) if they cannot happen at the same time. For example, when rolling a die, "rolling a 2" and "rolling a 5" are mutually exclusive — you cannot roll both on a single toss. But "rolling an even number" and "rolling a number greater than 3" are NOT mutually exclusive, because rolling a 4 satisfies both.

8.5 The Complement of an Event

The complement of event A, written as A' or A^c, is the event "A does not occur." Since either A happens or A does not happen (and these two possibilities cover everything), their probabilities must add up to 1.

P(A') = 1 - P(A)

[EXAM] Using the complement rule

The probability that a student passes STA1610 is 0.72. What is the probability that the student does NOT pass?

Step 1: Identify the given probability.

P(pass) = 0.72

Step 2: Apply the complement rule.

P(not pass) = 1 − P(pass)

= 1 − 0.72

= 0.28

P(not pass) = 0.28

Exam Tip: The complement rule is extremely useful when a question asks for the probability of "at least one" occurrence. Calculating P(at least one) directly can be complicated, but P(at least one) = 1 − P(none), and P(none) is usually much easier to find. STA1610 exams test this trick regularly.

8.6 Counting Techniques for Probability

When the sample space is large, listing all outcomes is impractical. We use counting rules to determine n(S) and n(E) efficiently.

8.6.1 The Multiplication Rule of Counting

If an experiment consists of k steps, and step 1 can occur in n_1 ways, step 2 in n_2 ways, and so on, then the total number of outcomes is $n_1 \times n_2 \times \ldots \times n_k$.

[CONCEPT] Multiplication counting

A restaurant offers 3 starters, 5 main courses, and 4 desserts. How many different three-course meals are possible?

Step 1: Multiply the choices at each stage.

Total meals = $3 \times 5 \times 4 = 60$

60 different meals

8.6.2 Factorials, Permutations, and Combinations

Factorial: $n! = n \times (n-1) \times (n-2) \times \square \times 2 \times 1$, and $0! = 1$ by definition.

Permutations (order matters): ${}^{n}P_{r} = n! / ((n-r)!)$

Combinations (order does NOT matter): ${}^{n}C_{r} = C(n, r) = n! / (r!(n-r)!)$

When to use which? Ask yourself: does the order of selection matter? If you are choosing a president, vice-president, and treasurer from a group of 10 people, the order matters (Alice as president is different from Alice as treasurer) — use permutations. If you are choosing a committee of 3 people from 10, the order does not matter (Alice, Bob, Charlie is the same committee as Charlie, Alice, Bob) — use combinations.

[EXAM] Combinations in probability

A bag contains 6 red balls and 4 blue balls. Three balls are drawn at random without replacement. Find the probability that exactly 2 are red.

Step 1: Find the total number of ways to choose 3 balls from 10.

$n(S) = C(10, 3) = 10! / (3!(10-3)!) = 10! / (3! \times 7!)$

$= (10 \times 9 \times 8) / (3 \times 2 \times 1)$

$= 720 / 6$

$= 120$

Step 2: Find the number of ways to choose exactly 2 red from 6 AND 1 blue from 4.

Ways to choose 2 red from 6: $C(6, 2) = 6! / (2! \times 4!) = (6 \times 5) / (2 \times 1) = 15$

Ways to choose 1 blue from 4: $C(4, 1) = 4$

By the multiplication rule: n(E) = 15 × 4 = 60

Step 3: Calculate the probability.

P(2 red) = 60 ÷ 120 = 0.5

P(exactly 2 red) = 0.5

TOPIC 09

Probability Rules

Addition rule, multiplication rule, conditional probability, and independence

9.1 The Addition Rule (for "OR" events)

When a question asks for the probability of A **or** B occurring, you use the addition rule. The key question is: can A and B happen at the same time?

9.1.1 Mutually Exclusive Events (Cannot Happen Together)

If A and B are mutually exclusive:
P(A or B) = P(A) + P(B)

[CONCEPT] **Addition rule for mutually exclusive events**

A die is rolled. Find the probability of rolling a 2 or a 5.

Step 1: Check mutual exclusivity.

You cannot roll a 2 and a 5 simultaneously on a single die → mutually exclusive □

Step 2: Add the individual probabilities.

P(2) = 1/6, P(5) = 1/6

P(2 or 5) = 1/6 + 1/6 = 2/6 = 1/3

P(2 or 5) = 1/3 ≈ 0.3333

9.1.2 Non-Mutually Exclusive Events (Can Happen Together)

If A and B can occur simultaneously, simply adding P(A) + P(B) would count the overlap twice. We must subtract the overlap once to correct for this double-counting.

General Addition Rule
P(A or B) = P(A) + P(B) - P(A and B)

Why subtract? Imagine a class of 100 students. 60 study Statistics (A), 50 study Accounting (B), and 20 study both. If you add 60 + 50 = 110, you have counted the 20 double-enrolled students twice. Subtracting 20 gives the correct count: 60 + 50 − 20 = 90 students study at least one of the two subjects.

[EXAM] General addition rule

In a group of 200 students, 120 own a laptop (L), 85 own a tablet (T), and 45 own both. Find the probability that a randomly selected student owns a laptop or a tablet.

Step 1: Calculate individual probabilities.

P(L) = 120 ÷ 200 = 0.60

P(T) = 85 ÷ 200 = 0.425

P(L and T) = 45 ÷ 200 = 0.225

Step 2: Apply the general addition rule.

P(L or T) = P(L) + P(T) − P(L and T)

= 0.60 + 0.425 − 0.225

= 1.025 − 0.225

= 0.80

P(laptop or tablet) = 0.80

9.2 Conditional Probability

Sometimes we want to know the probability of an event given that another event has already occurred. This is called **conditional probability**. The occurrence of the first event changes (or "conditions") the sample space for the second event.

P(A | B) = P(A and B) / P(B) Read as "the probability of A given B." The vertical bar "|" means "given that."

What does "given" really do? When you are told "given that B has occurred," you are shrinking the sample space from S down to just B. You are only looking at the outcomes inside B, and asking what fraction of those also belong to A. That is why we divide by P(B) — it becomes the new "total."

[EXAM] Conditional probability from a contingency table

A survey of 150 employees recorded their department and whether they support a new policy:

SupportOpposeTotal Sales402060 IT302555 HR201535 **Total9060150**

Find: (a) P(Support | Sales), (b) P(IT | Oppose).

Step 1: Part (a) — P(Support | Sales).

"Given Sales" means we restrict our attention to the Sales row only (60 employees).

Of those 60 Sales employees, 40 support the policy.

P(Support | Sales) = 40 ÷ 60 = 2/3 ≈ 0.6667

Step 2: Part (b) — P(IT | Oppose).

"Given Oppose" means we restrict to the Oppose column only (60 employees).

Of those 60 who oppose, 25 are from IT.

P(IT | Oppose) = 25 ÷ 60 = 5/12 ≈ 0.4167

P(Support | Sales) = 2/3, P(IT | Oppose) = 5/12

Verification using the formula:

P(Support and Sales) = 40 ÷ 150 = 4/15

P(Sales) = 60 ÷ 150 = 2/5

P(Support | Sales) = (4/15) ÷ (2/5) = (4/15) × (5/2) = 20/30 = 2/3 □

9.3 The Multiplication Rule (for "AND" events)

When a question asks for the probability of A **and** B both occurring, you use the multiplication rule. The form depends on whether the events are independent or dependent.

9.3.1 Independent Events

Two events are **independent** if the occurrence of one does not affect the probability of the other. For example, flipping a coin and rolling a die are independent — the coin result does not change the die probabilities.

If A and B are independent:
P(A and B) = P(A) × P(B)

[CONCEPT] Multiplication rule for independent events

A coin is tossed and a die is rolled. Find the probability of getting heads AND a 6.

Step 1: Confirm independence.

The coin toss does not affect the die roll → independent □

Step 2: Find individual probabilities.

P(Heads) = 1/2

P(6) = 1/6

Step 3: Multiply.

P(Heads and 6) = (1/2) × (1/6) = 1/12 ≈ 0.0833

P(Heads and 6) = 1/12

9.3.2 Dependent Events

If the occurrence of A changes the probability of B, the events are dependent. The multiplication rule then uses conditional probability.

General Multiplication Rule
P(A and B) = P(A) × P(B | A)

[EXAM] Multiplication rule for dependent events (without replacement)

A box contains 8 red marbles and 5 green marbles. Two marbles are drawn without replacement. Find the probability that both are red.

Step 1: Identify the events.

A = first marble is red, B = second marble is red

These are dependent because drawing the first marble changes the contents of the box.

Step 2: Find P(A).

Total marbles = 8 + 5 = 13

P(first red) = 8 ÷ 13

Step 3: Find P(B | A).

Given the first marble was red, the box now has 7 red and 5 green = 12 marbles total.

P(second red | first red) = 7 ÷ 12

Step 4: Multiply.

P(both red) = (8/13) × (7/12)

= 56 ÷ 156

= 14/39

≈ 0.3590

P(both red) = 14/39 ≈ 0.359

Common Exam Mistake: Students often treat "without replacement" problems as if they were independent and multiply (8/13) × (8/13). This is wrong. "Without replacement" always means dependent events — the denominator AND the numerator change after the first draw. "With replacement" means independent events.

9.4 Testing for Independence

How do you know whether two events are independent? You can test mathematically using the following rule:

A and B are independent if and only if:
P(A and B) = P(A) × P(B) Equivalently, A and B are independent if P(A | B) = P(A) (knowing B occurred does not change the probability of A).

[EXAM] Testing independence from a contingency table

Using the employee survey table from earlier, test whether "Support" and "Sales" are independent.

Step 1: Calculate P(Support and Sales).

P(Support and Sales) = 40 ÷ 150 = 0.2667

Step 2: Calculate P(Support) × P(Sales).

P(Support) = 90 ÷ 150 = 0.60

P(Sales) = 60 ÷ 150 = 0.40

P(Support) × P(Sales) = 0.60 × 0.40 = 0.24

Step 3: Compare.

0.2667 ≠ 0.24

Since P(Support and Sales) ≠ P(Support) × P(Sales), the events are NOT independent.

"Support" and "Sales" are dependent events. Being in the Sales department affects the probability of supporting the policy.

9.5 Summary of Probability Rules

Question Type	Rule	Condition
P(A or B)	P(A) + P(B) − P(A and B)	General (always works)
P(A or B)	P(A) + P(B)	A, B mutually exclusive only
P(A and B)	P(A) × P(B | A)	General (always works)
P(A and B)	P(A) × P(B)	A, B independent only
P(A | B)	P(A and B) ÷ P(B)	P(B) > 0
P(A')	1 − P(A)	Always

Exam Tip: STA1610 probability questions almost always involve contingency tables or "without replacement" scenarios. Master reading two-way tables and be very careful about whether the question says "with" or "without" replacement. Also, when you see "at least one," immediately think of the complement: P(at least one) = 1 − P(none).

TOPIC 10

Bayes' Theorem

Reversing conditional probability — updating beliefs with new evidence

10.1 Why Do We Need Bayes' Theorem?

In Topic 9, you learned how to calculate P(A | B) — the probability of A given that B has occurred. But sometimes you know P(B | A) and you need to work backwards to find P(A | B). These two conditional probabilities are *not* the same, and confusing them is one of the most common errors in statistics.

Real-world example: A medical test for a disease is 99% accurate. You test positive. What is the probability you actually have the disease? Your instinct might say 99%, but that is P(positive test | disease). What you actually want is P(disease | positive test), which could be much lower if the disease is rare. Bayes' Theorem lets you flip the conditional and get the correct answer.

10.2 The Law of Total Probability

Before stating Bayes' Theorem, we need a building block called the Law of Total Probability. If an event B can happen through several mutually exclusive "routes" (A_1, A_2, ..., A_k), then the total probability of B is the sum of the probabilities along each route.

Law of Total Probability
If A_1, A_2, ..., A_k are mutually exclusive and exhaustive (they cover the entire sample space), then:
$P(B) = P(A_1) \cdot P(B \mid A_1) + P(A_2) \cdot P(B \mid A_2) + \square + P(A_k) \cdot P(B \mid A_k)$

Think of it as a tree diagram. Each branch represents a different route to event B. You multiply along each branch (to get the joint probability) and then add across all branches (to get the total probability of B).

10.3 Bayes' Theorem

Bayes' Theorem (Two Events)
$P(A \mid B) = (P(A) \cdot P(B \mid A)) / P(B)$ where $P(B)$ is found using the Law of Total Probability:
$P(B) = P(A) \cdot P(B \mid A) + P(A') \cdot P(B \mid A')$

Bayes' Theorem (General Form with k events)
$P(A_i \mid B) = (P(A_i) \cdot P(B \mid A_i)) / (\Sigma_{j=1}^{k} P(A_j) \cdot P(B \mid A_j))$

The numerator is the probability of the specific route you are interested in. The denominator is the total probability of B across all possible routes. So Bayes' Theorem is really just asking: "Of all the ways B could have happened, what fraction came through route A?"

[EXAM] Bayes' Theorem with two events (medical testing)

A disease affects 2% of the population. A test for the disease has a 95% true positive rate (it correctly identifies 95% of sick people) and a 5% false positive rate (it incorrectly flags 5% of healthy people as sick). A person tests positive. What is the probability they actually have the disease?

Step 1: Define the events and write down the given probabilities.

D = person has the disease, D' = person does not have the disease

T = test is positive

P(D) = 0.02 (2% of population has the disease)

P(D') = 1 − 0.02 = 0.98

P(T | D) = 0.95 (true positive rate)

P(T | D') = 0.05 (false positive rate)

Step 2: State what we want to find.

We want P(D | T) — the probability of having the disease given a positive test.

Step 3: Calculate the numerator: P(D) × P(T | D).

= 0.02 × 0.95

= 0.019

Step 4: Calculate the denominator using the Law of Total Probability.

P(T) = P(D) × P(T | D) + P(D') × P(T | D')

= (0.02 × 0.95) + (0.98 × 0.05)

= 0.019 + 0.049

= 0.068

Step 5: Apply Bayes' Theorem.

P(D | T) = 0.019 ÷ 0.068

= 0.2794

P(disease | positive test) ≈ 0.279 or 27.9%. Despite the test being 95% accurate, a positive result only means about a 28% chance of actually having the disease. This is because the disease is rare (2%), so most positive results come from the much larger group of healthy people being falsely flagged.

[EXAM] Bayes' Theorem with three events (factory production)

A factory has three machines that produce items. Machine A produces 50% of all items, Machine B produces 30%, and Machine C produces 20%. The defect rates are: Machine A = 2%, Machine B = 3%, Machine C = 5%. An item is selected at random and found to be defective. What is the probability it came from Machine C?

Step 1: Define events and probabilities.

A, B, C = item produced by Machine A, B, C respectively

D = item is defective

P(A) = 0.50, P(B) = 0.30, P(C) = 0.20

P(D | A) = 0.02, P(D | B) = 0.03, P(D | C) = 0.05

Step 2: We want P(C | D).

Step 3: Calculate the numerator: P(C) × P(D | C).

= 0.20 × 0.05

= 0.010

Step 4: Calculate the denominator (total probability of a defective item).

P(D) = P(A)×P(D|A) + P(B)×P(D|B) + P(C)×P(D|C)

= (0.50 × 0.02) + (0.30 × 0.03) + (0.20 × 0.05)

= 0.010 + 0.009 + 0.010

= 0.029

Step 5: Apply Bayes' Theorem.

P(C | D) = 0.010 ÷ 0.029

= 0.3448

P(Machine C | defective) ≈ 0.345 or 34.5%. Even though Machine C produces only 20% of items, it accounts for about 34.5% of all defects because its defect rate is the highest.

Exam Tip: STA1610 Bayes' Theorem questions almost always follow the same structure: you are given prior probabilities (P(A), P(B), etc.) and conditional probabilities (P(evidence | A), P(evidence | B), etc.), and you must find the posterior probability (P(A | evidence)). Draw a tree diagram if it helps you organise the information. The numerator is always the single branch you care about; the denominator is the sum of all branches leading to the observed evidence.

Common Exam Mistake: Students confuse P(A | B) with P(B | A). These are fundamentally different quantities. P(positive test | disease) = 0.95 is NOT the same as P(disease | positive test) = 0.279. Always read the question carefully to identify which direction the conditional goes.

TOPIC 23

Discrete Random Variables

Probability distributions, expected value and variance—the missing exam topic identified in 2024 Q9 and 2019 Q11.

1. Probability Distribution

A discrete random variable X takes countable values. Its probability mass function must satisfy:

> **$0 \le P(X = x) \le 1$** for every x, and **$\Sigma P(X = x) = 1$**.

2. Expected Value (Mean)

The expected value is the probability-weighted long-run average:

> $\mu = E[X] = \Sigma\, xP(x)$

It does not have to equal any observable value of X.

3. Variance and Standard Deviation

> $\sigma^2 = \mathrm{Var}(X) = \Sigma(x-\mu)^2P(x) = \Sigma x^2P(x) - \mu^2$
> $\sigma = \sqrt{\mathrm{Var}(X)}$

[2024 Q9 · P.7] **Worked example**

X takes 0, 1, 2, 3, 4 with probabilities 0.3, 0.2, 0.2, 0.2, 0.1.

Step 1 — Check the distribution

> $0.3 + 0.2 + 0.2 + 0.2 + 0.1 = 1$.

Step 2 — Expected value

> $E[X] = 0(0.3) + 1(0.2) + 2(0.2) + 3(0.2) + 4(0.1) =$ **1.60**.

Step 3 — Second moment

> $E[X^2] = 0^2(0.3) + 1^2(0.2) + 2^2(0.2) + 3^2(0.2) + 4^2(0.1) = 4.40$.

Step 4 — Variance and standard deviation

> $\mathrm{Var}(X) = 4.40 - 1.60^2 =$ **1.84**; $\sigma = \sqrt{1.84} \approx$ **1.3565**.

> Exam answer for 2024 Q9: E[X] = 1.60. A useful extra check: Var(X) = 1.84.

Common trap: E[X] and Var(X) are different quantities. On this exact dataset, the expected value is 1.60 while the variance is 1.84—both values appear among the original options.

Open the 2024 source at page 7 ↗

TOPIC 11

The Binomial Distribution

Counting successes in a fixed number of yes/no trials

11.1 What Is a Binomial Experiment?

Many real-world situations involve repeating the same experiment a fixed number of times, where each repetition has only two possible outcomes: "success" or "failure." The binomial distribution gives us a formula to calculate the probability of getting exactly a certain number of successes.

Four Conditions for a Binomial Experiment
1. Fixed number of trials (n): The experiment is repeated exactly n times.
2. Two outcomes per trial: Each trial results in either "success" (S) or "failure" (F).
3. Constant probability of success (p): The probability of success is the same for every trial.
4. Independent trials: The outcome of one trial does not affect the outcome of any other trial.

Examples that ARE binomial: Flipping a coin 10 times (success = heads, n = 10, p = 0.5). Surveying 20 people about whether they own a car (success = yes, n = 20, p = some proportion). Testing 15 light bulbs for defects (success = defective, n = 15, p = defect rate).

Examples that are NOT binomial: Drawing cards without replacement (p changes after each draw, violating condition 3). Counting the number of customers arriving at a shop per hour (no fixed n, violating condition 1 — this is Poisson instead).

11.2 The Binomial Probability Formula

$P(X = x) = C(n, x)\ p^x\ (1-p)^{n-x}$ where:
X = the random variable representing the number of successes
x = the specific number of successes you want (0, 1, 2, ..., n)
n = total number of trials
p = probability of success on a single trial
1 - p = probability of failure (often written as q)
$C(n, x) = n! / (x!(n-x)!)$ = the number of ways to choose which x trials are successes

Breaking down the formula: The formula has three parts, each with a clear meaning:
• C(n, x): "In how many different arrangements can we get x successes out of n trials?" For example, if n = 3 and x = 1, the success could occur on trial 1, 2, or 3 — that is 3 arrangements.
• p^x: The probability of getting x successes (multiply p by itself x times).
• $(1-p)^{n-x}$: The probability of getting (n − x) failures (multiply the failure probability by itself the remaining number of times).
Multiplying all three gives the probability of exactly x successes in any order.

[CONCEPT] Basic binomial probability

A fair coin is tossed 5 times. Find the probability of getting exactly 3 heads.

Step 1: Identify the parameters.

> n = 5 (five tosses)

x = 3 (we want exactly 3 heads)

p = 0.5 (probability of heads on each toss)

1 − p = 0.5

Step 2: Calculate the combination C(5, 3).

C(5, 3) = 5! / (3!(5-3)!) = 5! / (3! × 2!)

= (5 × 4 × 3 × 2 × 1) / (3 × 2 × 1)(2 × 1)

= 120 / (6 × 2)

= 120 / 12

= 10

Step 3: Calculate p^x.

$0.5^3 = 0.5 \times 0.5 \times 0.5 = 0.125$

Step 4: Calculate $(1-p)^{n-x}$.

$0.5^{5-3} = 0.5^2 = 0.25$

Step 5: Multiply all three parts.

P(X = 3) = 10 × 0.125 × 0.25

= 10 × 0.03125

= 0.3125

P(exactly 3 heads) = 0.3125 (or 31.25%)

[EXAM] Binomial with unequal probabilities

Historically, 30% of students at UNISA fail STA1610 on their first attempt. If 8 students are selected at random, find the probability that (a) exactly 2 fail, (b) at most 1 fails.

Step 1: Identify parameters.

n = 8, p = 0.30 (probability of failure), 1 − p = 0.70

"Success" in the binomial sense = failing the exam (this is the event we are counting)

Step 2: Part (a) — P(X = 2).

C(8, 2) = 8! / (2! × 6!) = (8 × 7) / (2 × 1) = 28

$p^2 = 0.30^2 = 0.09$

$(1-p)^6 = 0.70^6$

$0.70^2 = 0.49$

$0.70^3 = 0.49 \times 0.70 = 0.343$

$0.70^6 = 0.343 \times 0.343 = 0.117649$

$P(X = 2) = 28 \times 0.09 \times 0.117649$

$= 28 \times 0.01058841$

$= 0.2965$

Step 3: Part (b) — $P(X \leq 1) = P(X = 0) + P(X = 1)$.

P(X = 0):

$C(8, 0) = 1$

$0.30^0 = 1$

$0.70^8 = 0.70^6 \times 0.70^2 = 0.117649 \times 0.49 = 0.05764801$

$P(X = 0) = 1 \times 1 \times 0.05764801 = 0.0576$

P(X = 1):

$C(8, 1) = 8$

$0.30^1 = 0.30$

$0.70^7 = 0.70^6 \times 0.70 = 0.117649 \times 0.70 = 0.0823543$

$P(X = 1) = 8 \times 0.30 \times 0.0823543$

$= 8 \times 0.02470629$

$= 0.1977$

P(X ≤ 1) = 0.0576 + 0.1977 = 0.2553

(a) P(exactly 2 fail) ≈ 0.2965, (b) P(at most 1 fails) ≈ 0.2553

11.3 Mean and Variance of the Binomial Distribution

Mean: $\mu = np$
Variance: $\sigma^2 = np(1-p)$
Standard Deviation: $\sigma = \sqrt{(np(1-p))}$

[CONCEPT] Binomial mean and standard deviation

Using the exam failure example (n = 8, p = 0.30), find the expected number of failures and the standard deviation.

Step 1: Mean.

$\mu = 8 \times 0.30 = 2.4$

On average, we expect 2.4 out of 8 students to fail.

Step 2: Variance.

$\sigma^2 = 8 \times 0.30 \times 0.70 = 1.68$

Step 3: Standard deviation.

$\sigma = \sqrt{1.68} = 1.2961$

$\mu = 2.4, \sigma \approx 1.30$

11.4 Binomial Probability Tables

In the STA1610 exam, you may be given binomial probability tables (or expected to use a calculator). The tables typically list $P(X \leq x)$ for various values of n and p. If the table gives cumulative probabilities, you can find individual probabilities by subtraction:

$P(X = x) = P(X \leq x) - P(X \leq x - 1)$

Exam Tip: STA1610 exams frequently ask binomial questions with phrases like "at least," "at most," "more than," or "fewer than." Translate these carefully:
• "At most 3" → $P(X \leq 3)$
• "At least 3" → $P(X \geq 3) = 1 - P(X \leq 2)$
• "More than 3" → $P(X > 3) = 1 - P(X \leq 3)$
• "Fewer than 3" → $P(X < 3) = P(X \leq 2)$
The most common mistake is an off-by-one error in these translations.

TOPIC 12

The Poisson Distribution

Counting rare events over a fixed interval of time, space, or volume

12.1 When Do We Use the Poisson Distribution?

The binomial distribution counts successes in a fixed number of trials. The Poisson distribution counts the number of times an event occurs in a fixed interval of time, space, area, or volume — when there is no fixed upper limit on the count and the events happen independently at a known average rate.

Examples of Poisson situations:
• The number of emails you receive per hour.
• The number of cars passing a toll gate per minute.
• The number of printing errors per page in a textbook.
• The number of customers arriving at a bank per 15-minute interval.
Notice that in each case, there is no fixed "number of trials." You cannot say there are exactly n opportunities for an email to arrive. The events just happen at some average rate.

Conditions for a Poisson Process
1. Events occur independently of each other.
2. The average rate of occurrence (λ) is constant over the interval.
3. Two events cannot occur at exactly the same instant (the probability of two simultaneous events is negligible).
4. The probability of an event in a very small sub-interval is proportional to the length of that sub-interval.

12.2 The Poisson Probability Formula

$P(X = x) = (e^{-\lambda} \cdot \lambda^{x}) / x!$ where:
X = the random variable (number of occurrences)
x = the specific number of occurrences (0, 1, 2, 3, ...)
λ (Greek letter "lambda") = the average (mean) number of occurrences per interval
e ≈ 2.71828 = Euler's number (a mathematical constant, like π)
x! = x factorial

Understanding λ: Lambda is the heart of the Poisson distribution. It represents the expected number of events per interval. If a call centre receives an average of 4 calls per minute, then λ = 4 for a one-minute interval. If you want the probability for a 5-minute interval, you must scale λ: λ = 4 × 5 = 20 for five minutes. Always make sure λ matches the interval specified in the question.

[CONCEPT] Basic Poisson probability

A hospital emergency room receives an average of 3 patients per hour between midnight and 6am. Find the probability that exactly 5 patients arrive in a given hour during this period.

Step 1: Identify the parameters.

> λ = 3 (average patients per hour)

x = 5 (we want exactly 5 patients)

Step 2: Calculate $e^{-\lambda}$.

$e^{-3} = 1 / (e^3)$

$e^3 = 2.71828^3$

$2.71828^2 = 7.38905$

$7.38905 \times 2.71828 = 20.0855$

$e^{-3} = 1 \div 20.0855 = 0.049787$

Step 3: Calculate λ^x.

$3^5 = 3 \times 3 \times 3 \times 3 \times 3 = 243$

Step 4: Calculate x!.

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Step 5: Substitute into the formula.

$P(X = 5) = (0.049787 \times 243) / 120$

$= 12.0982 / 120$

$= 0.1008$

P(exactly 5 patients) ≈ 0.1008 (or about 10.1%)

[EXAM] Poisson with interval scaling

A printing company finds an average of 2 errors per page. Find the probability that a 3-page report contains (a) exactly 4 errors, (b) no errors at all.

Step 1: Scale λ to match the interval.

Average rate = 2 errors per page

For a 3-page report: $\lambda = 2 \times 3 = 6$ errors per 3 pages

Step 2: Part (a) — P(X = 4) with λ = 6.

$e^{-6} = 0.002479$ (using calculator or tables)

$6^4 = 6 \times 6 \times 6 \times 6 = 1\ 296$

$4! = 24$

$P(X = 4) = (0.002479 \times 1\ 296) / 24$

= 3.2128 / 24

= 0.1339

Step 3: Part (b) — P(X = 0) with λ = 6.

$6^0 = 1$ (anything to the power 0 is 1)

0! = 1 (by definition)

$P(X = 0) = (e^{-6} \times 1) / 1 = e^{-6}$

= 0.002479

(a) P(exactly 4 errors) ≈ 0.1339, (b) P(no errors) ≈ 0.0025 (very unlikely!)

12.3 Mean and Variance of the Poisson Distribution

The Poisson distribution has a unique and elegant property: its mean and variance are equal.

Mean: $\mu = \lambda$
Variance: $\sigma^2 = \lambda$
Standard Deviation: $\sigma = \sqrt{\lambda}$

[CONCEPT] Poisson mean and standard deviation

If the average number of accidents at an intersection is 4 per month, state the mean, variance, and standard deviation.

Step 1: Identify λ.

$\lambda = 4$

Step 2: State the measures.

Mean = λ = 4 accidents per month

Variance = λ = 4

Standard deviation = √4 = 2 accidents per month

$\mu = 4$, $\sigma^2 = 4$, $\sigma = 2$

12.4 Poisson as an Approximation to the Binomial

When n is large and p is small, the binomial distribution becomes very close to a Poisson distribution with λ = np. This is useful because the Poisson formula is simpler to compute than the binomial formula when n is large.

Poisson Approximation to Binomial
Use when $n \geq 20$ and $p \leq 0.05$ (some textbooks say $n \geq 50$ and $np \leq 5$).
Set $\lambda = np$ and use the Poisson formula instead of the binomial formula.

[EXAM] Poisson approximation to binomial

A factory produces 1 000 items per day. The probability that any single item is defective is 0.003. Find the probability that exactly 5 items are defective in a day.

Step 1: Check if the Poisson approximation is appropriate.

$n = 1\ 000$ (large, ≥ 20 ✓)

$p = 0.003$ (small, ≤ 0.05 ✓)

Poisson approximation is appropriate.

Step 2: Calculate λ.

$\lambda = np = 1\ 000 \times 0.003 = 3$

Step 3: Apply the Poisson formula with $x = 5$, $\lambda = 3$.

$P(X = 5) = (e^{-3} \times 3^5) / 5!$

$= (0.049787 \times 243) / 120$

$= 12.0982 / 120$

$= 0.1008$

$P(\text{exactly 5 defective}) \approx 0.1008$

12.5 Binomial vs. Poisson — When to Use Which?

Feature	Binomial	Poisson
What is counted?	Successes in n fixed trials	Events in a continuous interval
Fixed number of trials?	Yes (n is fixed)	No (no upper limit)
Outcomes per trial?	Two (success/failure)	Count (0, 1, 2, 3, ...)
Key parameter(s)	n and p	λ (average rate)
Mean	np	λ
Variance	$np(1-p)$	λ
Typical keywords	"n items," "each with probability p"	"per hour," "per page," "per km"

Exam Tip: STA1610 exams test your ability to identify which distribution to use. Look for the keywords: if the question gives you a fixed number of trials and a probability of success per trial, it is binomial. If the question gives you an average rate per time/space interval, it is Poisson. If n is very large and p is very small, you may need the Poisson approximation to the binomial.

Common Exam Mistake: Forgetting to scale λ when the interval changes. If the average is 3 calls per hour and the question asks about a 2-hour period, you must use λ = 6, not λ = 3. Always check that λ matches the exact interval described in the question.

TOPIC 13

The Normal Distribution

The most important distribution in statistics — the bell curve

13.1 What Is the Normal Distribution?

The normal distribution is a continuous probability distribution that produces the famous symmetric "bell-shaped" curve. It is the single most important distribution in all of statistics because an enormous number of real-world variables follow it approximately: heights, weights, IQ scores, exam marks, measurement errors, and many more.

Even when the original data is not normally distributed, the normal distribution still plays a central role because of the Central Limit Theorem (Topic 14), which guarantees that sample means tend toward normality.

Properties of the Normal Curve

- **Bell-shaped and symmetric** about the mean. The left half is a mirror image of the right half.
- **Mean = Median = Mode.** All three measures of central tendency coincide at the centre of the curve.
- **The total area under the curve equals 1.** This is true of all probability distributions — the total probability must be 100%.
- **The curve extends infinitely in both directions** (from $-\infty$ to $+\infty$), but the tails get very close to zero very quickly.
- **The shape is completely determined by two parameters:** the mean (μ), which sets the centre, and the standard deviation (σ), which sets the spread.

The Empirical Rule (68-95-99.7 Rule)
For any normal distribution:
• Approximately 68% of the data falls within 1 standard deviation of the mean: $\mu \pm 1\sigma$
• Approximately 95% falls within 2 standard deviations: $\mu \pm 2\sigma$
• Approximately 99.7% falls within 3 standard deviations: $\mu \pm 3\sigma$

Why does this matter? The empirical rule gives you a quick sanity check. If exam marks are normally distributed with $\mu = 60$ and $\sigma = 10$, then about 68% of students scored between 50 and 70, about 95% scored between 40 and 80, and almost everyone (99.7%) scored between 30 and 90. If someone tells you that 40% of students scored below 30, you immediately know something is wrong — that contradicts the 99.7% rule.

13.2 The Standard Normal Distribution (Z)

There are infinitely many normal distributions — one for every possible combination of μ and σ. A normal distribution with μ = 170 and σ = 8 (heights of men in cm) looks different from one with μ = 60 and σ = 15 (exam marks). To avoid needing a separate probability table for every possible μ and σ, statisticians created a single "universal" normal distribution called the **standard normal distribution**.

> Standard Normal Distribution
> A normal distribution with μ = 0 and σ = 1.
> The random variable is denoted by Z (instead of X).
> Z ~ N(0, 1)

All normal probability calculations are done by first converting the original variable X into a Z-score, then looking up the Z-score in a standard normal table (the "Z-table"). This means you only ever need one table.

13.3 Calculating Z-Scores

A Z-score tells you how many standard deviations a value is above or below the mean. It standardises any normal distribution into the standard normal.

> Z-Score Formula
> z = (x - μ) / σ where x = the raw value, μ = the population mean, and σ = the population standard deviation.
>
> A positive z means the value is above the mean. A negative z means it is below the mean. A z of 0 means the value equals the mean.

[CONCEPT] Calculating z-scores

Exam marks are normally distributed with μ = 65 and σ = 12. Calculate the z-scores for marks of (a) 89, (b) 53, (c) 65.

Step 1: Part (a) — x = 89.

> z = (89 − 65) ÷ 12

> = 24 ÷ 12

> = 2.00

> Interpretation: A mark of 89 is exactly 2 standard deviations above the mean.

Step 2: Part (b) — x = 53.

> z = (53 − 65) ÷ 12

> = −12 ÷ 12

> = −1.00

> Interpretation: A mark of 53 is exactly 1 standard deviation below the mean.

Step 3: Part (c) — x = 65.

z = (65 − 65) ÷ 12

= 0 ÷ 12

= 0.00

Interpretation: A mark of 65 is exactly at the mean.

z(89) = 2.00, z(53) = −1.00, z(65) = 0.00

13.4 Finding Probabilities Using the Z-Table

The Z-table (found in your STA1610 study guide or exam formula sheet) gives the area under the standard normal curve to the **left** of a given z-value. This area represents the probability P(Z < z).

How to read the Z-table: The rows represent the z-value to one decimal place (e.g. 1.2), and the columns represent the second decimal place (e.g. 0.04). To find P(Z < 1.24), go to the row labelled 1.2 and the column labelled 0.04. The intersection gives the probability. Most Z-tables only show positive z-values; for negative z-values, use the symmetry of the normal curve (explained below).

13.4.1 Finding P(Z < z) — "Less Than"

This is the most direct use of the table. The table gives you this probability immediately.

[CONCEPT] P(Z < z) from the table

Find P(Z < 1.53).

Step 1: Locate the value in the Z-table.

Row = 1.5, Column = 0.03

Intersection = 0.9370

P(Z < 1.53) = 0.9370

13.4.2 Finding P(Z > z) — "Greater Than"

The table gives the area to the left. The area to the right is the complement.

P(Z gt; z) = 1 - P(Z lt; z)

[CONCEPT] P(Z > z)

Find P(Z > 1.53).

Step 1: Find P(Z < 1.53) from the table.

P(Z < 1.53) = 0.9370

Step 2: Subtract from 1.

$P(Z > 1.53) = 1 - 0.9370 = 0.0630$

$P(Z > 1.53) = 0.0630$

13.4.3 Finding P(a < Z < b) — "Between"

To find the area between two z-values, subtract the smaller cumulative area from the larger one.

P(a lt; Z lt; b) = P(Z lt; b) - P(Z lt; a)

[CONCEPT] P(a < Z < b)

Find $P(-1.20 < Z < 1.85)$.

Step 1: Find $P(Z < 1.85)$.

Row 1.8, Column 0.05 → 0.9678

Step 2: Find $P(Z < -1.20)$ using symmetry.

The normal curve is symmetric, so $P(Z < -1.20) = P(Z > 1.20)$

$P(Z > 1.20) = 1 - P(Z < 1.20)$

$P(Z < 1.20)$ from table: Row 1.2, Column 0.00 → 0.8849

$P(Z > 1.20) = 1 - 0.8849 = 0.1151$

Therefore $P(Z < -1.20) = 0.1151$

Step 3: Subtract.

$P(-1.20 < Z < 1.85) = 0.9678 - 0.1151 = 0.8527$

$P(-1.20 < Z < 1.85) = 0.8527$

13.4.4 Handling Negative Z-Values — The Symmetry Rule

Symmetry Properties of the Standard Normal
P(Z lt; -z) = P(Z gt; z) = 1 - P(Z lt; z)
P(-z lt; Z lt; z) = 2 × P(Z lt; z) - 1

13.5 Full Normal Probability Problems (X → Z → Probability)

In exam questions, you are usually given a normal variable X with a specific mean and standard deviation, and asked to find a probability. The procedure is always the same three steps: (1) sketch the curve, (2) convert X to Z, (3) look up the Z-table.

[EXAM] Normal probability — "less than"

The lifespan of a certain brand of light bulb is normally distributed with a mean of 800 hours and a standard deviation of 50 hours. Find the probability that a randomly selected bulb lasts less than 720 hours.

Step 1: Identify the parameters and the question.

$X \sim N(800, 50^2)$, so $\mu = 800$, $\sigma = 50$

We want $P(X < 720)$

Step 2: Convert to a z-score.

$z = (x - \mu) \div \sigma$

$= (720 - 800) \div 50$

$= -80 \div 50$

$= -1.60$

Step 3: Look up the probability.

$P(X < 720) = P(Z < -1.60)$

Using symmetry: $P(Z < -1.60) = 1 - P(Z < 1.60)$

From the Z-table: $P(Z < 1.60) = 0.9452$

$P(Z < -1.60) = 1 - 0.9452 = 0.0548$

$P(X < 720) = 0.0548$. There is about a 5.5% chance a bulb lasts less than 720 hours.

[EXAM] Normal probability — "between two values"

Using the same light bulb data ($\mu = 800$, $\sigma = 50$), find the probability that a bulb lasts between 750 and 870 hours.

Step 1: Convert both values to z-scores.

For $x = 750$: $z_1 = (750 - 800) \div 50 = -50 \div 50 = -1.00$

For $x = 870$: $z_2 = (870 - 800) \div 50 = 70 \div 50 = 1.40$

Step 2: Find the cumulative probabilities.

$P(Z < 1.40)$ from table: Row 1.4, Column 0.00 → 0.9192

$P(Z < -1.00) = 1 - P(Z < 1.00) = 1 - 0.8413 = 0.1587$

Step 3: Subtract.

$P(-1.00 < Z < 1.40) = 0.9192 - 0.1587 = 0.7605$

P(750 < X < 870) = 0.7605. About 76% of bulbs last between 750 and 870 hours.

[EXAM] Normal probability — "more than"

Find the probability that a bulb lasts more than 900 hours.

Step 1: Convert to z.

z = (900 − 800) ÷ 50 = 100 ÷ 50 = 2.00

Step 2: Find P(Z > 2.00).

P(Z > 2.00) = 1 − P(Z < 2.00)

From table: P(Z < 2.00) = 0.9772

P(Z > 2.00) = 1 − 0.9772 = 0.0228

P(X > 900) = 0.0228. Only about 2.3% of bulbs last more than 900 hours.

13.6 Finding X-Values from Probabilities (Reverse Lookup)

Sometimes the question gives you a probability and asks you to find the corresponding raw score. This is the reverse process: you look up the probability in the body of the Z-table to find the z-value, then convert back to X.

Converting Z back to X
x = μ + zσ This is simply the z-score formula rearranged to solve for x.

[EXAM] Finding a cutoff value from a probability

STA1610 exam marks are normally distributed with μ = 58 and σ = 14. The top 10% of students receive a distinction. What is the minimum mark needed for a distinction?

Step 1: Translate the question into a probability statement.

"Top 10%" means we want the value x such that P(X > x) = 0.10.

Equivalently, P(X < x) = 1 − 0.10 = 0.90.

Step 2: Find the z-value corresponding to a cumulative probability of 0.90.

Look in the body of the Z-table for the value closest to 0.9000.

P(Z < 1.28) = 0.8997 (very close to 0.90)

P(Z < 1.29) = 0.9015

The closest value is 1.28 (or more precisely, z ≈ 1.282 by interpolation).

We will use z = 1.28.

Step 3: Convert z back to x.

$x = \mu + z\sigma$

$= 58 + 1.28 \times 14$

$= 58 + 17.92$

$= 75.92$

A student needs approximately 76 marks (rounding up) to be in the top 10% and earn a distinction.

[EXAM] Finding the middle 80% range

Using the same exam data ($\mu = 58$, $\sigma = 14$), find the range of marks that contains the middle 80% of students.

Step 1: Determine the tail probabilities.

If the middle 80% is in the centre, then 20% is in the tails.

By symmetry, 10% is in each tail.

Lower cutoff: $P(X < x_1) = 0.10$

Upper cutoff: $P(X < x_2) = 0.90$

Step 2: Find the z-values.

For 0.90: $z = 1.28$ (from the previous example)

For 0.10: by symmetry, $z = -1.28$

Step 3: Convert both z-values to x.

$x_1 = 58 + (-1.28)(14) = 58 - 17.92 = 40.08$

$x_2 = 58 + (1.28)(14) = 58 + 17.92 = 75.92$

The middle 80% of students scored between approximately 40 and 76 marks.

13.7 Normal Approximation to the Binomial

When n is large, the binomial distribution starts to look like a normal distribution. This allows us to use the normal distribution (and Z-tables) to approximate binomial probabilities, which is much faster than calculating binomial coefficients for large n.

Conditions: Use the normal approximation when $np \geq 5$ and $n(1-p) \geq 5$.

Parameters: $\mu = np$, $\sigma = \sqrt{(np(1-p))}$

Continuity Correction: Because the binomial is discrete (counts) and the normal is continuous, we adjust by 0.5:
- $P(X = k) \approx P(k - 0.5 < X < k + 0.5)$
- $P(X \leq k) \approx P(X < k + 0.5)$
- $P(X \geq k) \approx P(X > k - 0.5)$

[EXAM] Normal approximation to binomial

In a large university, 40% of students own a car. A random sample of 100 students is selected. Use the normal approximation to find the probability that at least 45 own a car.

Step 1: Check conditions.

$n = 100$, $p = 0.40$

$np = 100 \times 0.40 = 40 \geq 5$ ☐

$n(1-p) = 100 \times 0.60 = 60 \geq 5$ ☐

Normal approximation is appropriate.

Step 2: Calculate μ and σ.

$\mu = np = 40$

$\sigma = \sqrt{(np(1-p))} = \sqrt{(100 \times 0.40 \times 0.60)} = \sqrt{24} = 4.899$

Step 3: Apply continuity correction.

$P(X \geq 45) \approx P(X > 44.5)$ (subtract 0.5 for "at least")

Step 4: Convert to z.

$z = (44.5 - 40) \div 4.899$

$= 4.5 \div 4.899$

$= 0.9186 \approx 0.92$

Step 5: Find the probability.

$P(Z > 0.92) = 1 - P(Z < 0.92)$

From table: $P(Z < 0.92) = 0.8212$

$P(Z > 0.92) = 1 - 0.8212 = 0.1788$

P(at least 45 own a car) ≈ 0.1788 or about 17.9%

Exam Tip: STA1610 exams test normal distribution more than any other single topic. You must be able to: (1) convert X to Z, (2) read the Z-table for "less than," "greater than," and "between" probabilities, (3) do reverse lookups to find X from a given probability, and (4) apply the normal approximation to the binomial with continuity correction. Practise all four types until they are automatic.

Common Exam Mistake: Forgetting the continuity correction when using the normal approximation to the binomial. If the question involves a discrete count (number of students, number of defective items, etc.) and you are using the normal approximation, you MUST apply the ±0.5 correction. Without it, your answer will be slightly off and you may lose marks.

TOPIC 14

Sampling Distributions & the Central Limit Theorem

Why sample means behave predictably even when individual data does not

14.1 The Big Idea: From One Sample to Many

So far, we have been working with individual data values. But in real statistics, we rarely care about a single observation. We care about *sample statistics* — the mean of a sample, the proportion of a sample, etc. — because we use those statistics to make inferences about the population.

Here is the key insight: if you take many random samples from the same population and calculate the mean of each sample, those sample means will vary from sample to sample. They form their own distribution, called the **sampling distribution of the sample mean**. Understanding this distribution is the foundation of all inferential statistics (confidence intervals and hypothesis testing in Topics 15–19).

Analogy: Imagine you want to know the average height of all UNISA students (the population mean μ). You cannot measure every student, so you take a random sample of 50 students and calculate their average height ($\bar{x}$). If you took a different sample of 50, you would get a slightly different $\bar{x}$. If you repeated this process thousands of times, the collection of all those $\bar{x}$ values would form a distribution. That distribution is the sampling distribution. The Central Limit Theorem tells us what this distribution looks like.

14.2 The Sampling Distribution of the Sample Mean

Properties of the Sampling Distribution of $\bar{X}$
If random samples of size n are drawn from a population with mean μ and standard deviation σ, then the sampling distribution of the sample mean $\bar{X}$ has:

Mean: $\mu_{\bar{x}} = \mu$ (the mean of the sample means equals the population mean)

Standard Deviation (Standard Error): $\sigma_{\bar{x}} = \sigma / \sqrt{(n)}$

The standard error measures how much sample means vary from sample to sample. Notice that as n increases, the standard error decreases — larger samples give more precise estimates.

Why is the standard error smaller than σ? Individual values can be far from the mean, but when you average many values together, the highs and lows tend to cancel out. The average of 100 values will be much closer to μ than any single value is. Dividing σ by √n quantifies this "averaging out" effect. For example, if σ = 20 and n = 100, the standard error is 20/√100 = 20/10 = 2. The sample means will typically be within about 2 units of μ, even though individual values can be 20 units away.

14.3 The Central Limit Theorem (CLT)

The Central Limit Theorem is arguably the most important theorem in all of statistics. It explains why the normal distribution appears so frequently in practice, even when the underlying population is not normal.

The Central Limit Theorem
If random samples of size n are drawn from any population with mean μ and finite standard deviation σ, then as n becomes large, the sampling distribution of the sample mean $\bar{X}$ approaches a normal distribution:
$\bar{X} \sim N(\mu, (\sigma^2) / n)$ regardless of the shape of the original population distribution.

Rule of thumb: The approximation is generally good when $n \geq 30$. If the population is already normal, the sampling distribution is exactly normal for any n (even n = 2).

Why is the CLT so powerful? It means you do not need to know the shape of the population distribution to make inferences about the mean. Whether the population is skewed, uniform, bimodal, or any other shape, as long as your sample size is large enough ($n \geq 30$), the sample means will be approximately normally distributed. This is what makes confidence intervals and hypothesis tests work in practice.

[CONCEPT] Understanding the CLT visually

Suppose the population distribution of monthly income in a town is heavily right-skewed (most people earn modest amounts, but a few earn very high incomes). Describe the sampling distribution of the mean for samples of size n = 50.

Step 1: Check the sample size.

> n = 50 ≥ 30, so the CLT applies.

Step 2: Describe the shape.

Even though the population is right-skewed, the sampling distribution of $\bar{X}$ will be approximately normal because n is large enough.

Step 3: Describe the centre and spread.

The mean of the sampling distribution = μ (the population mean income).

The standard error = $\sigma/\sqrt{50}$, which is much smaller than σ.

The sampling distribution is approximately $N(\mu, \sigma^2/50)$, despite the original population being skewed. This is the magic of the CLT.

14.4 Probability Calculations with the Sampling Distribution

Once we know the sampling distribution is normal (either because the population is normal or because $n \geq 30$ via the CLT), we can calculate probabilities about sample means using the same Z-score technique from Topic 13. The only difference is that we use the standard error instead of σ.

Z-Score for a Sample Mean
$z = (\bar{x} - \mu) / (\sigma / \sqrt{(n)})$ Compare this to the z-score for an individual value: $z = (x - \mu) / \sigma$. The only change is the denominator: σ becomes $\sigma/\sqrt{n}$.

[EXAM] Probability about a sample mean (population normal)

The weights of cereal boxes are normally distributed with $\mu = 500$g and $\sigma = 15$g. A random sample of 36 boxes is selected. Find the probability that the sample mean weight is less than 495g.

Step 1: Identify the parameters.

$\mu = 500$, $\sigma = 15$, $n = 36$

The population is normal, so the sampling distribution is exactly normal for any n.

Step 2: Calculate the standard error.

$\sigma_{\bar{x}} = \sigma / \sqrt{(n)} = 15 / \sqrt{(36)} = 15 / 6 = 2.5$

Step 3: Calculate the z-score for $\bar{x} = 495$.

$z = (495 - 500) / 2.5 = (-5) / 2.5 = -2.00$

Step 4: Find the probability.

$P(X < 495) = P(Z < -2.00)$

$= 1 - P(Z < 2.00)$

$= 1 - 0.9772$

$= 0.0228$

$P(\bar{X} < 495) = 0.0228$. There is only a 2.28% chance that the average weight of 36 boxes is below 495g, even though an individual box has a much higher chance of being below 495g.

[EXAM] Probability about a sample mean (CLT, non-normal population)

The time customers spend in a supermarket has a mean of 35 minutes and a standard deviation of 12 minutes. The distribution is right-skewed. A random sample of 64 customers is selected. Find the probability that the sample mean time is between 33 and 37 minutes.

Step 1: Check if the CLT applies.

$n = 64 \geq 30$ □. Even though the population is skewed, the sampling distribution of $\bar{X}$ is approximately normal.

Step 2: Calculate the standard error.

$\sigma_{\bar{x}} = 12 / \sqrt{(64)} = 12 / 8 = 1.5$

Step 3: Convert both bounds to z-scores.

For $\bar{x} = 33$: $z_1 = (33 - 35) \div 1.5 = -2 \div 1.5 = -1.33$

For $\bar{x} = 37$: $z_2 = (37 - 35) \div 1.5 = 2 \div 1.5 = 1.33$

Step 4: Find the probability.

$P(-1.33 < Z < 1.33) = P(Z < 1.33) - P(Z < -1.33)$

$P(Z < 1.33)$ from table: Row 1.3, Column 0.03 → 0.9082

$P(Z < -1.33) = 1 - 0.9082 = 0.0918$

$P(-1.33 < Z < 1.33) = 0.9082 - 0.0918 = 0.8164$

$P(33 < \bar{X} < 37) \approx 0.8164$. About 81.6% of samples of 64 customers will have a mean shopping time between 33 and 37 minutes.

Common Exam Mistake: Using σ instead of σ/√n in the z-score formula when the question asks about a sample mean. Read the question carefully: if it says "a randomly selected item" (individual), use σ. If it says "a random sample of n items" and asks about the sample mean, use σ/√n. This is the single most common error in STA1610 sampling distribution questions.

14.5 The Sampling Distribution of the Sample Proportion

Just as sample means have a sampling distribution, so do sample proportions. If you are interested in the proportion of a population with a certain characteristic (e.g. the proportion of voters who support a candidate), you estimate it with the sample proportion $\hat{p}$.

Sampling Distribution of $\hat{p}$
If random samples of size n are drawn from a population with proportion p, then for large n (specifically when $np \geq 5$ and $n(1-p) \geq 5$):

$\hat{p} \sim N(p, (p(1-p)) / n)$

Mean: $\mu_{\hat{p}} = p$
Standard Error: $\sigma_{\hat{p}} = \sqrt{((p(1-p)) / n)}$

Z-score for a sample proportion:
$z = (\hat{p} - p) / (\sqrt{((p(1-p)) / n)})$

[EXAM] Probability about a sample proportion

It is known that 35% of UNISA students are over the age of 30. A random sample of 200 students is selected. Find the probability that more than 40% of the sample are over 30.

Step 1: Check conditions.

$p = 0.35, n = 200$

$np = 200 \times 0.35 = 70 \geq 5$ □

$n(1-p) = 200 \times 0.65 = 130 \geq 5$ □

Normal approximation is appropriate.

Step 2: Calculate the standard error.

$\sigma_{\hat{p}} = \sqrt{((0.35 \times 0.65) / 200)}$

$= \sqrt{(0.2275 / 200)}$

$= \sqrt{(0.0011375)}$

$= 0.03373$

Step 3: Calculate the z-score for $\hat{p} = 0.40$.

$z = (0.40 - 0.35) \div 0.03373$

$= 0.05 \div 0.03373$

$= 1.4824 \approx 1.48$

Step 4: Find the probability.

$P(\hat{p} > 0.40) = P(Z > 1.48)$

$= 1 - P(Z < 1.48)$

From table: $P(Z < 1.48) = 0.9306$

$= 1 - 0.9306$

= 0.0694

P(p̂ > 0.40) ≈ 0.0694. There is about a 6.9% chance that more than 40% of a random sample of 200 students are over 30.

14.6 Finite Population Correction (FPC)

All the formulas above assume the population is very large compared to the sample (or that sampling is with replacement). If the sample size is more than 5% of the population (n/N > 0.05), the standard error should be adjusted with the finite population correction factor.

Finite Population Correction
$\sigma_{\bar{x}} = \sigma / \sqrt{n} \times \sqrt{((N - n) / (N - 1))}$ where N = population size. The correction factor $\sqrt{((N-n) / (N-1))}$ is always less than 1, so it reduces the standard error. In STA1610, you will rarely need this unless the question specifically gives you the population size N and n/N > 0.05.

Exam Tip: STA1610 sampling distribution questions fall into three categories:
1. Individual value: "Find the probability that a randomly selected item..." → use $z = (x - \mu)/\sigma$
2. Sample mean: "Find the probability that the sample mean..." → use $z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$
3. Sample proportion: "Find the probability that the sample proportion..." → use $z = (\hat{p} - p)/\sqrt{(p(1-p)/n)}$
The most common mistake is using the wrong formula. Always identify whether the question is about an individual, a mean, or a proportion before you start calculating.

Exam Tip: When the question says "the population is normally distributed," you can use the normal sampling distribution for any sample size n. When the question does NOT say the population is normal, you need n ≥ 30 to invoke the CLT. If n < 30 and the population is not stated to be normal, you cannot assume normality (you would need the t-distribution instead, covered in Topic 18).

TOPIC 15

Confidence Intervals

Estimating population parameters with a margin of error

15.1 Point Estimates vs. Interval Estimates

When we want to know a population parameter (like the true mean μ or the true proportion p), we use sample data to estimate it. There are two ways to do this:

- **Point estimate:** A single number. For example, "the sample mean is $\bar{x} = 72$, so we estimate $\mu \approx 72$." This is simple but gives no indication of how accurate the estimate is.
- **Interval estimate (confidence interval):** A range of values. For example, "we are 95% confident that μ is between 68 and 76." This is more informative because it tells you both the estimate and the uncertainty.

Analogy: Imagine someone asks you how far it is from Johannesburg to Cape Town. A point estimate is "1 400 km." A confidence interval is "between 1 350 and 1 450 km." The interval is more honest because it acknowledges that you are not exactly sure. The wider the interval, the less precise your estimate; the narrower, the more precise.

15.2 What Does "95% Confident" Actually Mean?

This is one of the most misunderstood concepts in statistics, and STA1610 exams test it regularly.

Correct Interpretation: If we were to take many random samples from the same population and construct a 95% confidence interval from each sample, then approximately 95% of those intervals would contain the true population parameter μ.

Incorrect Interpretation: "There is a 95% probability that μ is in this specific interval." This is wrong because μ is a fixed (but unknown) number — it is either in the interval or it is not. The 95% refers to the long-run success rate of the method, not to any single interval.

Think of it like a fishing net. A 95% confidence interval is like casting a net that catches the true fish (μ) 95% of the time. Once you have cast the net (calculated the interval), the fish is either in it or not — you just do not know which. But you trust the net because it works 95% of the time.

15.3 Confidence Interval for the Population Mean (σ Known)

When the population standard deviation σ is known (which is rare in practice but common in exam questions), we use the Z-distribution to construct the confidence interval.

CI for μ (σ known)
$\bar{x} \pm z_{\alpha/2} \cdot \sigma / \sqrt{n}$ where:
$\bar{x}$ = sample mean (the point estimate)
$z_{\alpha/2}$ = the critical z-value that leaves α/2 in each tail
σ = population standard deviation
n = sample size
$\sigma / \sqrt{n}$ = standard error
$z_{\alpha/2} \cdot \sigma / \sqrt{n}$ = margin of error (E)

Common Critical Z-Values

Confidence Level	α	α/2	zα/2
90%	0.10	0.05	1.645
95%	0.05	0.025	1.96
99%	0.01	0.005	2.576

These z-values come from the standard normal table. For a 95% confidence interval, you want 95% of the area in the centre, which leaves 5% in the tails — 2.5% in each tail. The z-value that has 2.5% above it (or equivalently, 97.5% below it) is 1.96.

[EXAM] 95% CI for the mean (σ known)

A random sample of 64 UNISA students has a mean study time of $\bar{x}$ = 18.5 hours per week. The population standard deviation is known to be σ = 6 hours. Construct a 95% confidence interval for the true mean study time of all UNISA students.

Step 1: Identify the given information.

$\bar{x}$ = 18.5, σ = 6, n = 64

Confidence level = 95%, so α = 0.05 and α/2 = 0.025

$z_{\alpha/2} = z_{0.025} = 1.96$

Step 2: Calculate the standard error.

$SE = \sigma \div \sqrt{n}$

$= 6 \div \sqrt{64}$

$= 6 \div 8$

$= 0.75$

Step 3: Calculate the margin of error (E).

$E = z_{\alpha/2} \times SE$

$= 1.96 \times 0.75$

$= 1.47$

Step 4: Construct the interval.

Lower limit = $\bar{x} - E = 18.5 - 1.47 = 17.03$

Upper limit = $\bar{x} + E = 18.5 + 1.47 = 19.97$

95% CI for μ: (17.03, 19.97) hours per week. We are 95% confident that the true mean study time of all UNISA students is between 17.03 and 19.97 hours per week.

[EXAM] 99% CI for the mean (σ known)

Using the same data ($\bar{x} = 18.5$, $\sigma = 6$, $n = 64$), construct a 99% confidence interval.

Step 1: Find the new critical value.

Confidence level = 99%, so $\alpha = 0.01$, $\alpha/2 = 0.005$

$z_{0.005} = 2.576$

Step 2: Calculate the new margin of error.

$E = 2.576 \times 0.75$

$= 1.932$

Step 3: Construct the interval.

Lower limit = $18.5 - 1.932 = 16.568$

Upper limit = $18.5 + 1.932 = 20.432$

99% CI for μ: (16.57, 20.43) hours. Notice that the 99% interval is wider than the 95% interval (16.57–20.43 vs. 17.03–19.97). Higher confidence requires a wider interval — you have to cast a bigger net to be more sure of catching μ.

15.4 Confidence Interval for the Population Mean (σ Unknown)

In real life, σ is almost never known. When σ is unknown, we estimate it using the sample standard deviation s, and we switch from the Z-distribution to the **t-distribution**.

CI for μ (σ unknown)
$\bar{x} \pm t_{\alpha/2,\, n-1} \cdot s / \sqrt{(n)}$ where $t_{\alpha/2,\, n-1}$ is the critical value from the t-distribution with n - 1 degrees of freedom, and s is the sample standard deviation.

Why the t-distribution? When we replace σ with s, we introduce extra uncertainty because s is itself an estimate that varies from sample to sample. The t-distribution accounts for this extra uncertainty by having heavier tails than the normal distribution. As n increases, s becomes a better estimate of σ, and the t-distribution converges to the normal distribution. For $n \geq 30$, the t and Z values are very close.

[EXAM] 95% CI for the mean (σ unknown, small sample)

A random sample of 16 employees yields a mean monthly overtime of $\bar{x} = 12.4$ hours with a sample standard deviation of $s = 3.8$ hours. Assuming overtime is normally distributed, construct a 95% confidence interval for the true mean overtime.

Step 1: Identify the given information.

$\bar{x} = 12.4$, $s = 3.8$, $n = 16$

σ is unknown → use the t-distribution

Degrees of freedom: $df = n - 1 = 16 - 1 = 15$

Confidence level = 95%, so $\alpha/2 = 0.025$

Step 2: Find the critical t-value.

From the t-table: $t_{0.025,\ 15} = 2.131$

(Compare this to $z_{0.025} = 1.96$ — the t-value is larger because of the extra uncertainty.)

Step 3: Calculate the standard error.

$SE = s \div \sqrt{n} = 3.8 \div \sqrt{16} = 3.8 \div 4 = 0.95$

Step 4: Calculate the margin of error.

$E = t \times SE = 2.131 \times 0.95$

$= 2.02445$

≈ 2.02

Step 5: Construct the interval.

Lower limit = $12.4 - 2.02 = 10.38$

Upper limit = $12.4 + 2.02 = 14.42$

95% CI for μ: (10.38, 14.42) hours. We are 95% confident that the true mean monthly overtime for all employees is between 10.38 and 14.42 hours.

15.5 Confidence Interval for a Population Proportion

When the parameter of interest is a proportion (e.g. the proportion of voters who support a candidate), we use the sample proportion $\hat{p}$ as the point estimate.

CI for p
$\hat{p} \pm z_{\alpha/2} \cdot \sqrt{((\hat{p}(1 - \hat{p})) / n)}$ Conditions: $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$ (to ensure the normal approximation is valid).

[EXAM] CI for a proportion

In a survey of 400 customers, 148 said they were satisfied with the service. Construct a 90% confidence interval for the true proportion of all customers who are satisfied.

Step 1: Calculate the sample proportion.

$\hat{p} = 148 \div 400 = 0.37$

$1 - \hat{p} = 1 - 0.37 = 0.63$

Step 2: Check conditions.

$n\hat{p} = 400 \times 0.37 = 148 \geq 5$ ✓

$n(1-\hat{p}) = 400 \times 0.63 = 252 \geq 5$ ✓

Step 3: Find the critical z-value.

90% confidence → $\alpha = 0.10$, $\alpha/2 = 0.05$

$z_{0.05} = 1.645$

Step 4: Calculate the standard error.

$SE = \sqrt{((0.37 \times 0.63) / 400)}$

$= \sqrt{(0.2331 / 400)}$

$= \sqrt{(0.00058275)}$

$= 0.02414$

Step 5: Calculate the margin of error.

$E = 1.645 \times 0.02414$

$= 0.03971$

≈ 0.0397

Step 6: Construct the interval.

Lower limit $= 0.37 - 0.0397 = 0.3303$

Upper limit $= 0.37 + 0.0397 = 0.4097$

90% CI for p: (0.330, 0.410). We are 90% confident that the true proportion of satisfied customers is between 33.0% and 41.0%.

15.6 Factors That Affect the Width of a Confidence Interval

The width of a confidence interval is 2E (twice the margin of error). Three factors control it:

Factor	Effect on Width	Why
Higher confidence level	Wider	Larger z or t critical value → larger E
Larger sample size (n)	Narrower	√n in the denominator → smaller SE → smaller E
Larger variability (σ or s)	Wider	Larger σ in the numerator → larger SE → larger E

Exam Tip: STA1610 frequently asks "How can we make the confidence interval narrower?" The three correct answers are: (1) decrease the confidence level, (2) increase the sample size, or (3) reduce the variability (which is usually not under your control). The most practical answer is to increase n.

15.7 Determining the Required Sample Size

Sometimes you need to plan a study and want to know how large a sample you need to achieve a desired margin of error at a given confidence level.

Sample Size for Estimating a Mean
$n = ((z_{\alpha/2} \cdot \sigma) / E)^2$ Always round UP to the next whole number (even if the calculation gives 96.1, you need n = 97).

Sample Size for Estimating a Proportion
$n = \hat{p}(1 - \hat{p})((z_{\alpha/2}) / E)^2$ If no prior estimate of $\hat{p}$ is available, use $\hat{p} = 0.5$ (this gives the maximum sample size, which is the most conservative choice).

[EXAM] Determining sample size for a mean

A researcher wants to estimate the mean monthly rent for apartments in Pretoria to within R200, with 95% confidence. A pilot study suggests σ ≈ R800. How many apartments must be sampled?

Step 1: Identify the values.

$E = 200$, $\sigma = 800$, $z_{0.025} = 1.96$

Step 2: Substitute into the formula.

$n = (1.96 \times 800 \div 200)^2$

$= (1\ 568 \div 200)^2$

$= (7.84)^2$

$= 61.4656$

Step 3: Round up.

n = 62 (always round up, even though 61.47 is closer to 61)

The researcher needs to sample at least 62 apartments.

Common Exam Mistake: Rounding the sample size down instead of up. If the formula gives n = 61.1, you need 62, not 61. Rounding down would give you a margin of error slightly larger than desired. Always round up to the next integer.

TOPIC 16

Hypothesis Testing Framework

The logic, language, and steps of statistical decision-making

16.1 What Is Hypothesis Testing?

A confidence interval gives you a range of plausible values for a population parameter. A hypothesis test goes a step further: it lets you make a formal decision about a specific claim regarding that parameter. For example, a manufacturer claims their light bulbs last an average of 1 000 hours. You suspect they last less. A hypothesis test provides a structured, evidence-based way to evaluate that claim.

Analogy: The Courtroom. Hypothesis testing works exactly like a criminal trial. The defendant is presumed innocent (the null hypothesis) until proven guilty (the alternative hypothesis). The prosecution presents evidence (the sample data). The jury decides whether the evidence is strong enough to reject the presumption of innocence. "Not guilty" does not mean the defendant is innocent — it just means the evidence was not strong enough to convict. Similarly, "fail to reject H_0" does not mean H_0 is true — it means the data did not provide enough evidence against it.

16.2 The Null and Alternative Hypotheses

Every hypothesis test begins with two competing statements about the population parameter:

Null Hypothesis (H_0): The "status quo" or "no effect" statement. It always contains an equality sign ($=$, $\leq$, or $\geq$). It is the hypothesis we assume to be true unless the data provides strong evidence against it.

Alternative Hypothesis (H_1 or H_a): The "research" or "claim" statement. It contains the inequality sign ($\neq$, $<$, or $>$). This is what we are trying to find evidence for.

Three Types of Tests

Test Type	H_0	H_1	When to Use
Two-tailed	$\mu = \mu_0$	$\mu \neq \mu_0$	Question asks "is there a difference?" or "has it changed?"
Left-tailed	$\mu \geq \mu_0$	$\mu < \mu_0$	Question asks "has it decreased?" or "is it less than?"
Right-tailed	$\mu \leq \mu_0$	$\mu > \mu_0$	Question asks "has it increased?" or "is it more than?"

How to identify the test type from the question: Look for keywords. "Different from," "changed," "not equal to" → two-tailed. "Less than," "decreased," "lower," "fewer" → left-tailed. "Greater than," "increased," "higher," "more" → right-tailed. The direction of the inequality in H_1 determines the tail.

[CONCEPT] Setting up hypotheses

Write the null and alternative hypotheses for each scenario:

(a) A cereal company claims their boxes contain an average of 500g. A consumer group suspects the boxes are underfilled.

H_0: $\mu \geq 500$ (the boxes contain at least 500g on average)

H_1: $\mu < 500$ (the boxes contain less than 500g — the consumer group's suspicion)

This is a left-tailed test.

(b) A lecturer believes the average exam mark has changed from the historical average of 62%.

H_0: $\mu = 62$ (the mean is still 62%)

H_1: $\mu \neq 62$ (the mean has changed — could be higher or lower)

This is a two-tailed test.

(c) A new teaching method is claimed to increase the pass rate above the current 65%.

H_0: $p \leq 0.65$ (the pass rate is at most 65%)

H_1: $p > 0.65$ (the pass rate has increased)

This is a right-tailed test.

16.3 The Test Statistic

The test statistic is a single number calculated from the sample data that measures how far the sample result is from what the null hypothesis predicts. It is essentially a z-score (or t-score) that tells you how many standard errors the sample statistic is away from the hypothesised parameter value.

General Form of a Test Statistic
Test Statistic = (Sample Statistic - Hypothesised Parameter) / Standard Error This has the same structure as a z-score. A large absolute value of the test statistic means the sample result is far from what H_0 predicts, providing evidence against H_0.

16.4 The Significance Level (α)

Before collecting data, you must decide how much evidence is "enough" to reject H_0. This threshold is the **significance level**, denoted by α (alpha).

Common Significance Levels
α = 0.10 (10% — weak evidence required)
α = 0.05 (5% — standard, most common in STA1610)
α = 0.01 (1% — strong evidence required)

The significance level is the probability of making a Type I error (rejecting H_0 when it is actually true). Choosing α = 0.05 means you are willing to accept a 5% chance of wrongly rejecting a true null hypothesis.

16.5 Two Approaches to Making the Decision

There are two equivalent ways to decide whether to reject H_0. STA1610 exams may use either approach, so you must know both.

16.5.1 The Critical Value Approach (Classical Approach)

Compare the test statistic to a critical value from the Z or t table. The critical value marks the boundary of the "rejection region."

Decision Rule (Critical Value Approach)
• Two-tailed test: Reject H_0 if |test statistic| > critical value (i.e. the test statistic falls in either tail).
• Left-tailed test: Reject H_0 if test statistic < −critical value.
• Right-tailed test: Reject H_0 if test statistic > +critical value.
Otherwise, do not reject H_0.

16.5.2 The P-Value Approach

The p-value is the probability of obtaining a test statistic as extreme as (or more extreme than) the one observed, assuming H_0 is true. It measures the strength of the evidence against H_0.

Decision Rule (P-Value Approach)
• If p-value ≤ α → Reject H_0 (the evidence is strong enough).
• If p-value > α → Do not reject H_0 (the evidence is not strong enough).

How to think about the p-value: The p-value answers the question: "If H_0 were true, how surprising would our sample result be?" A very small p-value (e.g. 0.003) means the observed result would be extremely unlikely if H_0 were true — so H_0 is probably not true. A large p-value (e.g. 0.45) means the observed result is quite consistent with H_0 — so we have no reason to reject it.

[CONCEPT] Interpreting p-values

For each p-value below, state the decision at α = 0.05.

(a) p-value = 0.03

0.03 ≤ 0.05 → Reject H_0. There is sufficient evidence at the 5% level.

(b) p-value = 0.12

0.12 > 0.05 → Do not reject H_0. There is insufficient evidence at the 5% level.

(c) p-value = 0.001

> $0.001 \leq 0.05 \rightarrow$ Reject H_0. The evidence is very strong.

(d) p-value = 0.049

> $0.049 \leq 0.05 \rightarrow$ Reject H_0 (just barely!). This is a borderline case.

16.6 Type I and Type II Errors

Because hypothesis testing is based on sample data (not the entire population), there is always a chance of making the wrong decision. There are two types of errors:

	H_0 is Actually True	H_0 is Actually False
Decision: Reject H_0	Type I Error (False Positive) Probability = α	Correct Decision Probability = $1 - \beta$ (Power)
Decision: Do Not Reject H_0	Correct Decision Probability = $1 - \alpha$	Type II Error (False Negative) Probability = β

Type I Error (α): Rejecting H_0 when H_0 is actually true. This is a "false alarm." The probability of a Type I error is exactly the significance level α that you chose.

Type II Error (β): Failing to reject H_0 when H_0 is actually false. This is a "missed detection." The probability of a Type II error is β, which depends on the true parameter value, sample size, and α.

Courtroom analogy continued: A Type I error is convicting an innocent person (rejecting the true presumption of innocence). A Type II error is acquitting a guilty person (failing to reject the presumption of innocence when the defendant is actually guilty). The justice system is designed to minimise Type I errors (better to let a guilty person go free than to convict an innocent one), which is why the burden of proof is high ("beyond reasonable doubt"). In statistics, we similarly control Type I error by choosing a small α.

Exam Tip: STA1610 exams frequently ask you to identify Type I and Type II errors in context. The key is to state what H_0 is, then describe what each error means in the specific scenario. For example, if H_0: "the new drug has no effect," then a Type I error is concluding the drug works when it actually does not, and a Type II error is concluding the drug does not work when it actually does.

16.7 The Complete Steps of Hypothesis Testing

Every hypothesis test follows the same structured procedure. Memorise these steps — you will use them in Topics 17, 18, 19, and 20.

The Six Steps of Hypothesis Testing

Step 1: State the null and alternative hypotheses (H_0 and H_1).
Step 2: Choose the significance level α.
Step 3: Identify the appropriate test statistic and its distribution (Z, t, χ^2, etc.).
Step 4: Calculate the test statistic from the sample data.
Step 5: Determine the critical value(s) or p-value, and make the decision.
Step 6: State the conclusion in the context of the original problem.

[EXAM] Full hypothesis test walkthrough (preview)

A manufacturer claims their batteries last an average of 500 hours. A consumer group tests 36 batteries and finds $\bar{x} = 490$ hours. The population standard deviation is known to be $\sigma = 30$ hours. Test the manufacturer's claim at $\alpha = 0.05$.

Step 1: State the hypotheses.

The consumer group suspects the batteries last less than claimed.

H_0: $\mu \geq 500$ (the manufacturer's claim is correct)

H_1: $\mu < 500$ (the batteries last less than claimed)

This is a left-tailed test.

Step 2: Choose α.

$\alpha = 0.05$ (given)

Step 3: Identify the test statistic.

σ is known and $n = 36 \geq 30 \rightarrow$ use the Z-test.

Test statistic: $z = (\bar{x} - \mu_0) / (\sigma / \sqrt{(n)})$

Step 4: Calculate the test statistic.

$z = (490 - 500) \div (30 \div \sqrt{36})$

$= -10 \div (30 \div 6)$

$= -10 \div 5$

$= -2.00$

Step 5: Make the decision.

Critical value approach:

For a left-tailed test at $\alpha = 0.05$, the critical value is $z_{crit} = -1.645$.

Since $-2.00 < -1.645$, the test statistic falls in the rejection region.

Decision: Reject H_0.

P-value approach (verification):

p-value = $P(Z < -2.00) = 0.0228$

Since $0.0228 < 0.05$, reject H_0. □ (Both approaches agree.)

Step 6: State the conclusion in context.

At the 5% significance level, there is sufficient evidence to conclude that the mean battery life is less than 500 hours. The manufacturer's claim is not supported by the data.

Reject H_0. The batteries last significantly less than the claimed 500 hours ($z = -2.00$, $p = 0.0228$).

16.8 Connecting Confidence Intervals and Hypothesis Tests

There is a direct relationship between two-tailed hypothesis tests and confidence intervals. For a two-tailed test at significance level α, you can use the $(1 - \alpha) \times 100\%$ confidence interval to make the decision:

CI–Hypothesis Test Connection
- If the hypothesised value μ_0 falls outside the $(1 - \alpha)$ confidence interval → Reject H_0.
- If μ_0 falls inside the confidence interval → Do not reject H_0.

This only works for two-tailed tests. For one-tailed tests, you must use the formal hypothesis testing procedure.

[CONCEPT] Using a CI to make a hypothesis test decision

A 95% confidence interval for the mean exam mark is (58.3, 67.1). Test H_0: $\mu = 70$ vs. H_1: $\mu \neq 70$ at $\alpha = 0.05$.

Step 1: Check if $\mu_0 = 70$ is inside the CI.

The CI is (58.3, 67.1).

70 is NOT inside this interval ($70 > 67.1$).

Step 2: Make the decision.

Since 70 falls outside the 95% CI, we reject H_0 at $\alpha = 0.05$.

Reject H_0. There is sufficient evidence that the mean exam mark is different from 70.

16.9 Summary of the Hypothesis Testing Framework

Concept	Key Point
H_0	Contains equality; assumed true until evidence says otherwise
H_1	Contains inequality; what you are trying to prove
α	Probability of Type I error; threshold for decision (usually 0.05)
Test statistic	Measures how far the sample result is from H_0 in standard error units
p-value	Probability of getting a result this extreme if H_0 is true
Decision rule	Reject H_0 if p-value $\leq \alpha$ (or if test statistic is in rejection region)
Type I error	Rejecting a true H_0 (false alarm); probability = α
Type II error	Failing to reject a false H_0 (missed detection); probability = β

Exam Tip: STA1610 hypothesis testing questions always follow the same pattern. The specific test statistic formula changes depending on the parameter (mean, proportion, etc.) and the conditions (σ known vs. unknown, large vs. small sample), but the six-step framework is always the same. Master the framework in this topic, and Topics 17–20 will simply be applications of it with different formulas plugged into Step 4.

Common Exam Mistake: Saying "accept H_0" instead of "do not reject H_0." In hypothesis testing, we never "accept" the null hypothesis. We can only say the evidence is insufficient to reject it. This is a subtle but important distinction that STA1610 markers look for. Think of the courtroom: "not guilty" is not the same as "innocent."

TOPIC 17

Z-Test for the Population Mean

Testing claims about μ when the population standard deviation σ is known

17.1 When to Use the Z-Test for a Mean

The Z-test for the population mean is used when you want to test a claim about the population mean μ and the following conditions are met:

- The population standard deviation σ is **known**.
- The population is normally distributed, OR the sample size is large ($n \geq 30$) so the Central Limit Theorem applies.
- The sample is a simple random sample.

How often is σ actually known? In real research, almost never. You usually have to estimate σ from the sample (using s), which means you would use the t-test instead (Topic 18). However, STA1610 exam questions frequently give you σ to test your ability to apply the Z-test. In the exam, if the question gives you σ explicitly, use the Z-test. If it gives you s (the sample standard deviation), use the t-test.

17.2 The Z-Test Statistic

Z-Test Statistic for a Mean
$z = (\bar{x} - \mu_0) / (\sigma / \sqrt{(n)})$ where:
$\bar{x}$ = sample mean
μ_0 = the hypothesised population mean (the value stated in H_0)
σ = known population standard deviation
n = sample size
$\sigma / \sqrt{(n)}$ = standard error of the mean

The test statistic measures how many standard errors the sample mean is away from the hypothesised mean. If this number is large (in absolute value), it means the sample result is far from what H_0 predicts, providing evidence against H_0.

17.3 Critical Values for the Z-Test

Test Type	α = 0.10	α = 0.05	α = 0.01
Two-tailed	±1.645	±1.96	±2.576
Left-tailed	−1.282	−1.645	−2.326
Right-tailed	+1.282	+1.645	+2.326

17.4 Worked Examples

[EXAM] Two-tailed Z-test for a mean

A university claims that the average time students spend on assignments per week is 10 hours. A researcher suspects this is no longer accurate. She surveys 49 randomly selected students and finds a sample mean of $\bar{x} = 11.2$ hours. The population standard deviation is known to be $\sigma = 3.5$ hours. Test the university's claim at the 5% significance level.

Step 1: State the hypotheses.

The researcher suspects the mean is "no longer accurate" — it could be higher or lower.

H_0: $\mu = 10$ (the university's claim)

H_1: $\mu \neq 10$ (the mean has changed)

This is a two-tailed test.

Step 2: Choose the significance level.

$\alpha = 0.05$ (given)

Step 3: Identify the test statistic.

σ is known (3.5), $n = 49 \geq 30 \rightarrow$ Z-test is appropriate.

$z = (\bar{x} - \mu_0) / (\sigma / \sqrt{(n)})$

Step 4: Calculate the test statistic.

$\bar{x} = 11.2$, $\mu_0 = 10$, $\sigma = 3.5$, $n = 49$

Standard error $= \sigma \div \sqrt{n} = 3.5 \div \sqrt{49} = 3.5 \div 7 = 0.5$

$z = (11.2 - 10) \div 0.5$

$= 1.2 \div 0.5$

$= 2.40$

Step 5: Make the decision.

Critical value approach:

Two-tailed test at $\alpha = 0.05 \rightarrow$ critical values $= \pm 1.96$

Since $|2.40| = 2.40 > 1.96$, the test statistic falls in the rejection region.

Decision: Reject H_0.

P-value approach:

p-value = 2 × P(Z > 2.40) (multiply by 2 because two-tailed)

P(Z > 2.40) = 1 − P(Z < 2.40) = 1 − 0.9918 = 0.0082

p-value = 2 × 0.0082 = 0.0164

Since 0.0164 < 0.05, reject H_0. □

Step 6: Conclusion in context.

At the 5% significance level, there is sufficient evidence to conclude that the mean time students spend on assignments per week is different from 10 hours. The university's claim is not supported by the data.

z = 2.40, p = 0.0164. Reject H_0 at α = 0.05.

[EXAM] Left-tailed Z-test for a mean

A coffee machine is designed to dispense 250 ml per cup. A quality inspector suspects the machine is underfilling. She measures 36 cups and finds $\bar{x}$ = 247 ml. The population standard deviation is σ = 8 ml. Test at α = 0.01 whether the machine is underfilling.

Step 1: State the hypotheses.

H_0: μ ≥ 250 (the machine dispenses at least 250 ml)

H_1: μ < 250 (the machine underfills)

Left-tailed test.

Step 2: α = 0.01.

Step 3: Test statistic — Z-test (σ known, n = 36 ≥ 30).

Step 4: Calculate.

SE = 8 ÷ √36 = 8 ÷ 6 = 1.3333

z = (247 − 250) ÷ 1.3333

= −3 ÷ 1.3333

= −2.25

Step 5: Decision.

Critical value: Left-tailed at α = 0.01 → $z_{crit} = -2.326$

Since −2.25 > −2.326, the test statistic does NOT fall in the rejection region.

Decision: Do not reject H_0.

P-value: $P(Z < -2.25) = 1 - P(Z < 2.25) = 1 - 0.9878 = 0.0122$

Since $0.0122 > 0.01$, do not reject H_0. ☐

Step 6: Conclusion.

At the 1% significance level, there is insufficient evidence to conclude that the machine is underfilling. Although the sample mean (247 ml) is below 250 ml, the difference is not statistically significant at this strict level.

$z = -2.25$, $p = 0.0122$. Do not reject H_0 at $\alpha = 0.01$.

[EXAM] Right-tailed Z-test for a mean

The average salary for entry-level accountants in South Africa is reported to be R18 000 per month. A recruitment agency believes the true mean is higher. They sample 81 accountants and find $\bar{x}$ = R19 200. The population standard deviation is σ = R4 500. Test at $\alpha = 0.05$.

Step 1: Hypotheses.

H_0: $\mu \leq 18\,000$

H_1: $\mu > 18\,000$ (right-tailed)

Step 2: $\alpha = 0.05$.

Step 3: Z-test (σ known, $n = 81 \geq 30$).

Step 4: Calculate.

$SE = 4\,500 \div \sqrt{81} = 4\,500 \div 9 = 500$

$z = (19\,200 - 18\,000) \div 500$

$= 1\,200 \div 500$

$= 2.40$

Step 5: Decision.

Critical value: $z_{0.05} = 1.645$ (right-tailed)

Since $2.40 > 1.645$, reject H_0.

P-value: $P(Z > 2.40) = 1 - 0.9918 = 0.0082$

$0.0082 < 0.05 \rightarrow$ reject H_0. ☐

Step 6: Conclusion.

At the 5% significance level, there is sufficient evidence to conclude that the mean entry-level accountant salary exceeds R18 000 per month.

z = 2.40, p = 0.0082. Reject H_0 at α = 0.05.

Exam Tip: STA1610 Z-test questions always give you σ explicitly. If you see σ in the question, use the Z-test. The calculation is straightforward: subtract the hypothesised mean from the sample mean, then divide by the standard error. The most common mistakes are (1) using s instead of σ in the denominator, (2) forgetting to divide σ by √n, and (3) using the wrong tail for the critical value.

TOPIC 18

T-Test for the Population Mean

Testing claims about μ when σ is unknown and must be estimated from the sample

18.1 When to Use the T-Test

The t-test is used when you want to test a claim about the population mean μ but the population standard deviation σ is **unknown**. Instead, you estimate it using the sample standard deviation s. Because s is an estimate (not the true value), there is extra uncertainty, and the t-distribution accounts for this by having heavier tails than the normal distribution.

Conditions for the T-Test
1. σ is unknown (you are given s instead).
2. The population is normally distributed, OR the sample size is large (n ≥ 30).
3. The sample is a simple random sample.

Important: When n is small (n < 30), the population MUST be approximately normal for the t-test to be valid. When n ≥ 30, the CLT ensures approximate normality regardless of the population shape.

18.2 The T-Test Statistic

T-Test Statistic for a Mean
$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$ with degrees of freedom: df = n - 1

Notice this is identical to the Z-test formula except that σ is replaced by s. The only other difference is that we compare the result to the t-distribution (using the t-table) instead of the Z-distribution.

Why does the t-distribution have heavier tails? When you use s instead of σ, your estimate of the standard error is less precise, especially with small samples. The heavier tails of the t-distribution reflect this extra uncertainty — they make it harder to reject H_0, which is the correct conservative behaviour when you have less information. As n increases, s becomes a better estimate of σ, and the t-distribution converges to the normal distribution. By the time n = 120, the t and Z critical values are nearly identical.

18.3 Reading the T-Table

The t-table is organised differently from the Z-table. The rows represent degrees of freedom (df = n − 1), and the columns represent the tail area (α for one-tailed tests, or α/2 for two-tailed tests).

For example, to find the critical t-value for a two-tailed test at α = 0.05 with n = 20:

- df = 20 − 1 = 19
- Column: α/2 = 0.025 (because it is two-tailed)
- $t_{0.025,\ 19} = 2.093$

> Common Exam Mistake: Using the wrong column in the t-table. For a two-tailed test at α = 0.05, you need the column labelled 0.025 (α/2), NOT 0.05. For a one-tailed test at α = 0.05, you use the column labelled 0.05. Always check whether your test is one-tailed or two-tailed before looking up the critical value.

18.4 Worked Examples

[EXAM] Two-tailed t-test (small sample)

A pharmaceutical company claims their painkiller provides relief for an average of 8 hours. A doctor tests the drug on 12 patients and records the following relief times (in hours): 7.2, 8.5, 6.8, 9.1, 7.5, 8.0, 7.8, 6.5, 8.3, 7.0, 9.0, 7.3. Test the company's claim at α = 0.05. Assume relief times are normally distributed.

Step 1: State the hypotheses.

> H_0: μ = 8 (the company's claim)

> H_1: μ ≠ 8 (two-tailed — we are testing whether the claim is accurate)

Step 2: α = 0.05.

Step 3: Identify the test statistic.

> σ is unknown (we only have the sample data), n = 12 < 30, but the population is stated to be normal → t-test is appropriate.

> df = 12 − 1 = 11

Step 4: Calculate the sample mean and standard deviation.

> **Sample mean:**

> $\Sigma x = 7.2 + 8.5 + 6.8 + 9.1 + 7.5 + 8.0 + 7.8 + 6.5 + 8.3 + 7.0 + 9.0 + 7.3$

> $= 93.0$

> $\bar{x} = 93.0 \div 12 = 7.75$

> **Sample standard deviation (using the computational formula):**

> $\Sigma x^2 = 7.2^2 + 8.5^2 + 6.8^2 + 9.1^2 + 7.5^2 + 8.0^2 + 7.8^2 + 6.5^2 + 8.3^2 + 7.0^2 + 9.0^2 + 7.3^2$

> $= 51.84 + 72.25 + 46.24 + 82.81 + 56.25 + 64.00 + 60.84 + 42.25 + 68.89 + 49.00 + 81.00 + 53.29$

= 728.66

$s^2 = (728.66 - (93.0^2) / 12) / (12 - 1)$

= (728.66 - 8 649 / 12) / 11

= (728.66 - 720.75) / 11

= 7.91 / 11

= 0.7191

s = √(0.7191) = 0.8480

Step 5: Calculate the test statistic.

SE = s ÷ √n = 0.8480 ÷ √12 = 0.8480 ÷ 3.4641 = 0.2448

t = (7.75 − 8) ÷ 0.2448

= −0.25 ÷ 0.2448

= −1.021

Step 6: Make the decision.

Critical values: two-tailed, α = 0.05, df = 11 → $t_{0.025,\ 11} = \pm 2.201$

Since |−1.021| = 1.021 < 2.201, the test statistic does NOT fall in the rejection region.

Decision: Do not reject H_0.

Step 7: Conclusion.

At the 5% significance level, there is insufficient evidence to conclude that the mean relief time is different from 8 hours. The company's claim is not contradicted by the data.

t = −1.021, df = 11. Do not reject H_0 at α = 0.05.

[EXAM] Right-tailed t-test (summary data given)

A fitness trainer claims that a new exercise programme increases the average number of push-ups a person can do beyond the national average of 25. After the programme, a sample of 20 participants can do an average of $\bar{x} = 28.5$ push-ups with $s = 6.2$. Test the claim at $\alpha = 0.05$. Assume push-up counts are approximately normal.

Step 1: Hypotheses.

H_0: $\mu \leq 25$

H_1: $\mu > 25$ (right-tailed — the trainer claims an increase)

Step 2: α = 0.05.

Step 3: Test statistic.

> σ unknown, n = 20, population approximately normal → t-test.

> df = 20 − 1 = 19

Step 4: Calculate.

> SE = 6.2 ÷ √20 = 6.2 ÷ 4.4721 = 1.3864

> t = (28.5 − 25) ÷ 1.3864

> = 3.5 ÷ 1.3864

> = 2.525

Step 5: Decision.

> Critical value: right-tailed, α = 0.05, df = 19 → $t_{0.05,\ 19} = 1.729$

> Since 2.525 > 1.729, reject H_0.

Step 6: Conclusion.

> At the 5% significance level, there is sufficient evidence to support the trainer's claim that the exercise programme increases the average number of push-ups beyond 25.

> t = 2.525, df = 19. Reject H_0 at α = 0.05.

[EXAM] Left-tailed t-test

A factory claims its light bulbs last an average of at least 1 200 hours. A consumer watchdog tests 16 bulbs and finds $\bar{x}$ = 1 150 hours with s = 80 hours. Test the claim at α = 0.05. Assume bulb lifespans are normally distributed.

Step 1: Hypotheses.

> H_0: μ ≥ 1 200 (the factory's claim)

> H_1: μ < 1 200 (left-tailed — suspecting shorter lifespan)

Step 2: α = 0.05.

Step 3: t-test (σ unknown, n = 16, population normal). df = 15.

Step 4: Calculate.

> SE = 80 ÷ √16 = 80 ÷ 4 = 20

> t = (1 150 − 1 200) ÷ 20

> = −50 ÷ 20

> = −2.50

Step 5: Decision.

> Critical value: left-tailed, $\alpha = 0.05$, $df = 15 \rightarrow t_{0.05,\,15} = -1.753$

> Since $-2.50 < -1.753$, reject H_0.

Step 6: Conclusion.

> At the 5% significance level, there is sufficient evidence to conclude that the mean lifespan of the bulbs is less than 1 200 hours. The factory's claim is not supported.

> $t = -2.50$, $df = 15$. Reject H_0 at $\alpha = 0.05$.

18.5 Z-Test vs. T-Test — Decision Flowchart

Condition	Use
σ known, any n (population normal or n ≥ 30)	Z-test
σ unknown, n ≥ 30	T-test (though Z is also acceptable as an approximation)
σ unknown, n < 30, population normal	T-test (must use t)
σ unknown, n < 30, population NOT normal	Neither — use non-parametric methods (beyond STA1610 scope)

Exam Tip: The simplest way to decide on the STA1610 exam: if the question gives you σ (population standard deviation), use Z. If it gives you s (sample standard deviation), use t. This rule works for virtually every STA1610 question. Also, when n is large (≥ 30), the t and Z critical values are very close, so even if you accidentally use the wrong one, your answer will be approximately correct. But for small samples, the difference matters — t critical values are larger than Z critical values, making it harder to reject H_0.

Common Exam Mistake: Using the Z-table instead of the t-table when σ is unknown and n is small. For example, with df = 10 and α = 0.05 (two-tailed), the t critical value is 2.228, but the Z critical value is 1.96. Using 1.96 would make it too easy to reject H_0 and could lead to a wrong conclusion.

TOPIC 19

Z-Test for a Population Proportion

Testing claims about percentages and proportions

19.1 When to Use the Z-Test for a Proportion

When the parameter of interest is a population proportion p (e.g. the proportion of voters who support a candidate, the proportion of defective items, the proportion of students who pass), we use a Z-test based on the sample proportion $\hat{p}$.

Conditions
1. The sample is a simple random sample.
2. The conditions for the binomial distribution are met (fixed n, two outcomes, independent trials, constant p).
3. The sample is large enough for the normal approximation: $np_0 \geq 5$ and $n(1 - p_0) \geq 5$, where p_0 is the hypothesised proportion from H_0.

19.2 The Z-Test Statistic for a Proportion

$z = (\hat{p} - p_0) / (\sqrt{((p_0(1 - p_0)) / n)})$ where:
$\hat{p} = x / n$ = sample proportion (x = number of successes in the sample)
p_0 = hypothesised population proportion (from H_0)
n = sample size

Important: The standard error uses p_0 (the hypothesised value), NOT $\hat{p}$ (the sample value). This is because the test statistic is calculated under the assumption that H_0 is true.

Why use p_0 instead of $\hat{p}$ in the denominator? In hypothesis testing, we assume H_0 is true when calculating the test statistic. If H_0 says p = 0.40, then under H_0 the standard error should be calculated using 0.40, not the sample proportion. This is different from confidence intervals (Topic 15), where we use $\hat{p}$ because there is no hypothesised value to plug in.

19.3 Worked Examples

[EXAM] Two-tailed Z-test for a proportion

A political analyst claims that exactly 50% of registered voters support the new tax policy. A survey of 400 randomly selected voters finds that 220 support the policy. Test the analyst's claim at α = 0.05.

Step 1: State the hypotheses.

H_0: p = 0.50 (the analyst's claim)

H_1: p ≠ 0.50 (two-tailed — testing whether the proportion is different from 50%)

Step 2: α = 0.05.

Step 3: Check conditions and identify the test statistic.

$n = 400$, $p_0 = 0.50$

$np_0 = 400 \times 0.50 = 200 \geq 5$ ✓

$n(1-p_0) = 400 \times 0.50 = 200 \geq 5$ ✓

Normal approximation is valid → Z-test for proportion.

Step 4: Calculate the test statistic.

$\hat{p} = 220 \div 400 = 0.55$

Standard error $= \sqrt{((p_0(1-p_0)) / n)} = \sqrt{((0.50 \times 0.50) / 400)}$

$= \sqrt{(0.25 / 400)}$

$= \sqrt{(0.000625)}$

$= 0.025$

$z = (0.55 - 0.50) \div 0.025$

$= 0.05 \div 0.025$

$= 2.00$

Step 5: Decision.

Critical value: Two-tailed at $\alpha = 0.05$ → $z_{crit} = \pm 1.96$

Since $|2.00| = 2.00 > 1.96$, reject H_0.

P-value: $2 \times P(Z > 2.00) = 2 \times (1 - 0.9772) = 2 \times 0.0228 = 0.0456$

$0.0456 < 0.05$ → reject H_0. ✓

Step 6: Conclusion.

At the 5% significance level, there is sufficient evidence to conclude that the proportion of voters who support the tax policy is different from 50%. The analyst's claim is not supported.

$z = 2.00$, $p = 0.0456$. Reject H_0 at $\alpha = 0.05$.

[EXAM] Left-tailed Z-test for a proportion

A mobile phone company claims that at least 80% of its customers are satisfied with their service. A consumer organisation surveys 250 customers and finds that 185 are satisfied. Test the company's claim at $\alpha = 0.01$.

Step 1: Hypotheses.

H_0: $p \geq 0.80$ (the company's claim)

> H_1: $p < 0.80$ (left-tailed — suspecting satisfaction is lower)

Step 2: $\alpha = 0.01$.

Step 3: Check conditions.

> $np_0 = 250 \times 0.80 = 200 \geq 5$ ✓

> $n(1-p_0) = 250 \times 0.20 = 50 \geq 5$ ✓

Step 4: Calculate.

> $\hat{p} = 185 \div 250 = 0.74$

> $SE = \sqrt{((0.80 \times 0.20) / 250)} = \sqrt{(0.16 / 250)} = \sqrt{(0.00064)} = 0.02530$

> $z = (0.74 - 0.80) \div 0.02530$

> $= -0.06 \div 0.02530$

> $= -2.372$

Step 5: Decision.

> Critical value: left-tailed at $\alpha = 0.01 \rightarrow z_{crit} = -2.326$

> Since $-2.372 < -2.326$, reject H_0.

> P-value: $P(Z < -2.37) = 1 - P(Z < 2.37) = 1 - 0.9911 = 0.0089$

> $0.0089 < 0.01 \rightarrow$ reject H_0. ✓

Step 6: Conclusion.

> At the 1% significance level, there is sufficient evidence to conclude that the proportion of satisfied customers is less than 80%. The company's claim is not supported.

> $z = -2.37$, $p = 0.0089$. Reject H_0 at $\alpha = 0.01$.

[EXAM] Right-tailed Z-test for a proportion

Historically, 30% of STA1610 students score above 75%. After introducing a new tutorial programme, a sample of 150 students shows that 54 scored above 75%. Test at $\alpha = 0.05$ whether the new programme has increased the proportion of high scorers.

Step 1: Hypotheses.

> H_0: $p \leq 0.30$

> H_1: $p > 0.30$ (right-tailed — testing for an increase)

Step 2: $\alpha = 0.05$.

Step 3: Check conditions.

$np_0 = 150 \times 0.30 = 45 \geq 5$ ✓

$n(1-p_0) = 150 \times 0.70 = 105 \geq 5$ ✓

Step 4: Calculate.

$\hat{p} = 54 \div 150 = 0.36$

$SE = \sqrt{((0.30 \times 0.70) / 150)} = \sqrt{(0.21 / 150)} = \sqrt{(0.0014)} = 0.03742$

$z = (0.36 - 0.30) \div 0.03742$

$= 0.06 \div 0.03742$

$= 1.603$

Step 5: Decision.

Critical value: right-tailed at $\alpha = 0.05 \rightarrow z_{crit} = 1.645$

Since $1.603 < 1.645$, do NOT reject H_0.

P-value: $P(Z > 1.60) = 1 - 0.9452 = 0.0548$

$0.0548 > 0.05 \rightarrow$ do not reject H_0. ✓

Step 6: Conclusion.

At the 5% significance level, there is insufficient evidence to conclude that the new tutorial programme has increased the proportion of students scoring above 75%. Although the sample proportion (36%) is higher than the historical 30%, the difference is not statistically significant at the 5% level.

$z = 1.60$, $p = 0.0548$. Do not reject H_0 at $\alpha = 0.05$.

19.4 Summary of All Hypothesis Tests for Means and Proportions

Parameter	Conditions	Test Statistic	Distribution
Mean μ	σ known	$z = (\bar{x} - \mu_0) / (\sigma/\sqrt{(n)})$	Standard Normal (Z)
Mean μ	σ unknown	$t = (\bar{x} - \mu_0) / (s/\sqrt{(n)})$	t with df = n−1
Proportion p	$np_0 \geq 5$, $n(1-p_0) \geq 5$	$z = (\hat{p} - p_0) / (\sqrt{(p_0(1-p_0)/n)})$	Standard Normal (Z)

Exam Tip: STA1610 hypothesis testing questions for means and proportions are very formulaic. The six steps never change. The only thing that changes is the formula in Step 4. Memorise the three formulas in the table above, and you can handle any hypothesis test question on the exam. Pay close attention to whether the question gives you σ or s (this determines Z vs. t) and whether the parameter is a mean or a proportion.

Common Exam Mistake: Using $\hat{p}$ instead of p_0 in the standard error of the proportion test. The standard error must use the hypothesised value p_0 because we are calculating the test statistic under the assumption that H_0 is true. Using $\hat{p}$ in the denominator is correct for confidence intervals but wrong for hypothesis tests. This is a very common error that costs students marks.

TOPIC 20

Chi-Square Tests

Testing relationships between categorical variables and goodness of fit

20.1 What Is the Chi-Square Distribution?

The chi-square (χ^2) distribution is a family of right-skewed distributions used for testing hypotheses about categorical data. Unlike the Z and t distributions, which deal with means and proportions, the chi-square distribution deals with **counts** and **frequencies**.

There are two main types of chi-square tests in STA1610:

- **Chi-Square Goodness-of-Fit Test:** Tests whether a single categorical variable follows a specified distribution (e.g. "Are the colours of sweets equally distributed?").
- **Chi-Square Test of Independence:** Tests whether two categorical variables are related (e.g. "Is gender related to voting preference?").

20.2 The Chi-Square Test Statistic

Both types of chi-square tests use the same formula. The idea is simple: compare the **observed** frequencies (what you actually counted in the data) to the **expected** frequencies (what you would expect if the null hypothesis were true). If the observed and expected frequencies are very different, the chi-square statistic will be large, providing evidence against H_0.

Chi-Square Test Statistic
$\chi^2 = \Sigma ((O - E)^2) / E$ where:
O = observed frequency (the actual count from the data)
E = expected frequency (the count predicted by H_0)
The sum is taken over all categories or cells in the table.

Why square the differences? The same reason we square deviations in the variance formula — to prevent positive and negative differences from cancelling out. Dividing by E standardises the squared difference, giving more weight to cells where the expected count is small (a difference of 5 is more surprising when E = 10 than when E = 1 000).

20.3 Chi-Square Goodness-of-Fit Test

This test checks whether the observed distribution of a single categorical variable matches a theoretical distribution.

Hypotheses
H_0: The data follows the specified distribution (the proportions are as claimed).
H_1: The data does not follow the specified distribution.

Expected Frequencies: $E_i = n \times p_i$, where p_i is the hypothesised proportion for category i.

Degrees of Freedom: df = k - 1, where k = number of categories.

Condition: All expected frequencies should be ≥ 5.

[EXAM] Goodness-of-fit test

A die is rolled 120 times. The observed frequencies for each face are:

Face123456 Observed (O)182215252020

Test at α = 0.05 whether the die is fair.

Step 1: State the hypotheses.

H_0: The die is fair (each face has probability 1/6).

H_1: The die is not fair (at least one face has a different probability).

Step 2: Calculate the expected frequencies.

If the die is fair, each face should appear n × p = 120 × (1/6) = 20 times.

E = 20 for all six faces.

Step 3: Calculate the chi-square statistic.

Face 1: $(18 - 20)^2 \div 20 = (-2)^2 \div 20 = 4 \div 20 = 0.20$

Face 2: $(22 - 20)^2 \div 20 = (2)^2 \div 20 = 4 \div 20 = 0.20$

Face 3: $(15 - 20)^2 \div 20 = (-5)^2 \div 20 = 25 \div 20 = 1.25$

Face 4: $(25 - 20)^2 \div 20 = (5)^2 \div 20 = 25 \div 20 = 1.25$

Face 5: $(20 - 20)^2 \div 20 = 0 \div 20 = 0.00$

Face 6: $(20 - 20)^2 \div 20 = 0 \div 20 = 0.00$

$\chi^2 = 0.20 + 0.20 + 1.25 + 1.25 + 0.00 + 0.00 = 2.90$

Step 4: Find the critical value.

df = k − 1 = 6 − 1 = 5

From the χ^2 table at α = 0.05, df = 5: $\chi^2_{crit} = 11.070$

Step 5: Decision.

Since 2.90 < 11.070, do not reject H_0.

Step 6: Conclusion.

At the 5% significance level, there is insufficient evidence to conclude that the die is unfair. The observed frequencies are consistent with a fair die.

$\chi^2 = 2.90$, df = 5, $\chi^2_{crit} = 11.070$. Do not reject H_0.

20.4 Chi-Square Test of Independence

This test determines whether two categorical variables are related (dependent) or unrelated (independent). The data is organised in a **contingency table** (a two-way table of counts).

Hypotheses
H_0: The two variables are independent (no relationship).
H_1: The two variables are dependent (there is a relationship).

Expected Frequency for Each Cell:
E = (Row Total) × (Column Total) / Grand Total

Degrees of Freedom: df = (r - 1)(c - 1), where r = number of rows and c = number of columns.

Condition: All expected frequencies should be ≥ 5 (some textbooks allow up to 20% of cells to have E ≥ 1 as long as all E ≥ 1).

[EXAM] Chi-square test of independence

A survey of 200 employees recorded their department and stress level:

Low StressHigh StressRow Total Sales305080 IT403070 HR351550 **Column Total10595200**

Test at α = 0.05 whether department and stress level are independent.

Step 1: Hypotheses.

H_0: Department and stress level are independent.

H_1: Department and stress level are dependent.

Step 2: Calculate expected frequencies for each cell.

E(Sales, Low) = (80 × 105) ÷ 200 = 8 400 ÷ 200 = 42.0

E(Sales, High) = (80 × 95) ÷ 200 = 7 600 ÷ 200 = 38.0

E(IT, Low) = (70 × 105) ÷ 200 = 7 350 ÷ 200 = 36.75

E(IT, High) = (70 × 95) ÷ 200 = 6 650 ÷ 200 = 33.25

E(HR, Low) = (50 × 105) ÷ 200 = 5 250 ÷ 200 = 26.25

E(HR, High) = (50 × 95) ÷ 200 = 4 750 ÷ 200 = 23.75

Step 3: Calculate $(O - E)^2 / E$ for each cell.

Sales, Low: $(30 - 42)^2 \div 42 = 144 \div 42 = 3.4286$

Sales, High: $(50 - 38)^2 \div 38 = 144 \div 38 = 3.7895$

IT, Low: $(40 - 36.75)^2 \div 36.75 = 10.5625 \div 36.75 = 0.2874$

IT, High: $(30 - 33.25)^2 \div 33.25 = 10.5625 \div 33.25 = 0.3177$

HR, Low: $(35 - 26.25)^2 \div 26.25 = 76.5625 \div 26.25 = 2.9167$

HR, High: $(15 - 23.75)^2 \div 23.75 = 76.5625 \div 23.75 = 3.2237$

Step 4: Sum to get χ^2.

$\chi^2 = 3.4286 + 3.7895 + 0.2874 + 0.3177 + 2.9167 + 3.2237$

$= 13.9636$

Step 5: Find the critical value.

$df = (r - 1)(c - 1) = (3 - 1)(2 - 1) = 2 \times 1 = 2$

From the χ^2 table at $\alpha = 0.05$, df = 2: $\chi^2_{crit} = 5.991$

Step 6: Decision.

Since $13.96 > 5.991$, reject H_0.

Step 7: Conclusion.

At the 5% significance level, there is sufficient evidence to conclude that department and stress level are dependent. The stress level distribution differs significantly across departments.

$\chi^2 = 13.96$, df = 2, $\chi^2_{crit} = 5.991$. Reject H_0.

Exam Tip: STA1610 chi-square questions are very mechanical. The hardest part is calculating the expected frequencies correctly. Remember: E = (row total × column total) ÷ grand total. Double-check that your expected frequencies sum to the same row and column totals as the observed frequencies. Also, the chi-square test is always right-tailed — you only reject H_0 when χ^2 is large (greater than the critical value). There is no left-tailed or two-tailed chi-square test.

Common Exam Mistake: Using the wrong degrees of freedom. For goodness-of-fit: df = k − 1 (number of categories minus 1). For independence: df = (r − 1)(c − 1). Students sometimes mix these up. Also, never use the chi-square test if any expected frequency is less than 5 — the approximation breaks down.

TOPIC 21

Correlation

Measuring the strength and direction of a linear relationship between two variables

21.1 What Is Correlation?

Correlation measures how closely two quantitative variables move together in a straight-line (linear) pattern. For example, as study hours increase, do exam marks tend to increase? As temperature rises, do ice cream sales tend to increase? Correlation quantifies this tendency with a single number between −1 and +1.

> Correlation does NOT imply causation. This is the most important warning in all of statistics. Just because two variables are correlated does not mean one causes the other. Ice cream sales and drowning deaths are positively correlated (both increase in summer), but eating ice cream does not cause drowning. A third variable (hot weather) drives both. Always be cautious about drawing causal conclusions from correlation alone.

21.2 The Pearson Correlation Coefficient (r)

> Pearson Correlation Coefficient
> $r = (n\Sigma xy - (\Sigma x)(\Sigma y)) / (\sqrt{([n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]})$ where:
> x and y = the two variables
> n = number of paired observations
> Σxy = sum of the products of each pair
> Σx, Σy = sums of each variable
> Σx^2, Σy^2 = sums of the squares

Properties of r

- **Range:** $-1 \leq r \leq +1$.
- **r = +1:** Perfect positive linear relationship (all points fall exactly on an upward-sloping line).
- **r = −1:** Perfect negative linear relationship (all points fall exactly on a downward-sloping line).
- **r = 0:** No linear relationship (the points show no straight-line pattern).
- **Sign of r:** Positive means both variables tend to increase together. Negative means as one increases, the other tends to decrease.
- **Magnitude of r:** The closer |r| is to 1, the stronger the linear relationship.

|r| Range	Strength
0.00 – 0.29	Weak
0.30 – 0.69	Moderate
0.70 – 1.00	Strong

21.3 Calculating r Step by Step

[EXAM] Calculating the correlation coefficient

The following data shows the number of hours studied (x) and the exam mark (y) for 6 students:

Studentx (hours)y (mark) 1245 2455 3565 4770 5880 61085

Step 1: Build the calculation table.

We need columns for x, y, xy, x^2, and y^2.

xyxyx²y² 2459042 025 455220163 025 565325254 225 770490494 900 880640646 400 10858501007 225 **Σx = 36Σy = 400Σxy = 2 615Σx² = 258Σy² = 27 800**

Step 2: Identify the sums.

n = 6, Σx = 36, Σy = 400, Σxy = 2 615, Σx^2 = 258, Σy^2 = 27 800

Step 3: Calculate the numerator.

$n\Sigma xy - (\Sigma x)(\Sigma y) = 6 \times 2\,615 - 36 \times 400$

$= 15\,690 - 14\,400$

$= 1\,290$

Step 4: Calculate the denominator.

First part: $n\Sigma x^2 - (\Sigma x)^2 = 6 \times 258 - 36^2 = 1\,548 - 1\,296 = 252$

Second part: $n\Sigma y^2 - (\Sigma y)^2 = 6 \times 27\,800 - 400^2 = 166\,800 - 160\,000 = 6\,800$

Denominator = $\sqrt{(252 \times 6\,800)} = \sqrt{1\,713\,600} = 1\,309.05$

Step 5: Calculate r.

$r = 1\,290 \div 1\,309.05 = 0.9855$

r ≈ 0.986. This indicates a very strong positive linear relationship between study hours and exam marks. As study hours increase, marks tend to increase in a nearly perfect straight-line pattern.

21.4 The Coefficient of Determination (r^2)

The coefficient of determination tells you what proportion of the variation in y is explained by the linear relationship with x.

$r^2 = (r)^2$ Expressed as a percentage, r^2 tells you the percentage of the total variation in y that is accounted for by the linear relationship with x.

[CONCEPT] Interpreting r^2

Using r = 0.986 from the previous example, calculate and interpret r^2.

Step 1: Square r.

$r^2 = (0.986)^2 = 0.9722$

Step 2: Interpret.

97.22% of the variation in exam marks can be explained by the linear relationship with study hours. The remaining 2.78% is due to other factors (natural ability, sleep, exam anxiety, etc.) or random variation.

$r^2 = 0.972$. The linear model explains 97.2% of the variation in marks.

Exam Tip: STA1610 exams frequently ask you to (a) calculate r from a data table, (b) interpret the strength and direction of the relationship, and (c) calculate and interpret r^2. Make sure you can build the calculation table quickly and accurately. Also, remember that r only measures LINEAR relationships. Two variables could have a strong curved relationship but $r \approx 0$.

Common Exam Mistake: Confusing r and r^2. If the question asks for the "correlation coefficient," give r. If it asks for the "coefficient of determination" or "the proportion of variation explained," give r^2. Also, r can be negative, but r^2 is always positive (squaring removes the sign).

21.5 Testing the Significance of r

A sample correlation r might be non-zero just by chance, even if the true population correlation ρ is zero. We can test whether r is statistically significant using a t-test.

Test Statistic for Correlation
$t = (r\sqrt{(n - 2)}) / (\sqrt{(1 - r^2)})$ with $df = n - 2$.

H_0: $\rho = 0$ (no linear correlation in the population)
H_1: $\rho \neq 0$ (there is a linear correlation)

[EXAM] Testing significance of r

Using the study hours data ($r = 0.986$, $n = 6$), test whether the correlation is significant at $\alpha = 0.05$.

Step 1: Hypotheses.

H_0: $\rho = 0$, H_1: $\rho \neq 0$ (two-tailed)

Step 2: Calculate the test statistic.

$t = (0.986 \times \sqrt{(6 - 2)}) \div \sqrt{(1 - 0.986^2)}$

$= (0.986 \times \sqrt{4}) \div \sqrt{(1 - 0.9722)}$

$= (0.986 \times 2) \div \sqrt{0.0278}$

$= 1.972 \div 0.1667$

$= 11.83$

Step 3: Critical value.

$df = 6 - 2 = 4$, $\alpha = 0.05$ (two-tailed) → $t_{crit} = \pm 2.776$

Step 4: Decision.

> Since |11.83| > 2.776, reject H_0.

> The correlation is statistically significant (t = 11.83, df = 4, p < 0.05). There is strong evidence of a linear relationship between study hours and exam marks in the population.

TOPIC 22

Simple Linear Regression

Finding the best-fitting straight line to predict one variable from another

22.1 What Is Regression?

Correlation tells you whether two variables are linearly related and how strongly. Regression goes a step further: it finds the actual equation of the straight line that best fits the data, allowing you to **predict** the value of one variable (the dependent variable, y) from the value of the other (the independent variable, x).

> Analogy: Correlation tells you "study hours and marks are strongly related." Regression tells you "for every additional hour of study, marks increase by about 5.2 points, and a student who studies zero hours would be expected to score about 35." The regression equation turns the relationship into a predictive tool.

22.2 The Regression Equation

> Simple Linear Regression Model
> $\hat{y} = a + bx$ where:
> $\hat{y}$ = predicted value of the dependent variable
> x = value of the independent variable
> b = slope of the regression line (the change in y for a one-unit increase in x)
> a = y-intercept (the predicted value of y when x = 0)

The "hat" on $\hat{y}$ indicates that this is a predicted (estimated) value, not the actual observed value. The actual values will scatter around the regression line due to random variation.

22.3 Calculating the Slope (b) and Intercept (a)

> Slope
> $b = (n\Sigma xy - (\Sigma x)(\Sigma y)) / (n\Sigma x^2 - (\Sigma x)^2)$ Notice that the numerator is the same as the numerator of the correlation coefficient r.
>
> Intercept
> $a = \bar{y} - b\bar{x} = \Sigma y / n - b \cdot \Sigma x / n$

[EXAM] Finding the regression equation

Using the study hours data from Topic 21, find the regression equation and predict the mark for a student who studies 6 hours.

From Topic 21: n = 6, $\Sigma x = 36$, $\Sigma y = 400$, $\Sigma xy = 2\,615$, $\Sigma x^2 = 258$, $\Sigma y^2 = 27\,800$

Step 1: Calculate the slope b.

Numerator = $n\Sigma xy - (\Sigma x)(\Sigma y) = 6 \times 2\,615 - 36 \times 400 = 15\,690 - 14\,400 = 1\,290$

Denominator = $n\Sigma x^2 - (\Sigma x)^2 = 6 \times 258 - 36^2 = 1\,548 - 1\,296 = 252$

$b = 1\,290 \div 252 = 5.1190$

Step 2: Calculate the intercept a.

$\bar{x} = 36 \div 6 = 6$

$\bar{y} = 400 \div 6 = 66.6667$

$a = \bar{y} - b\bar{x} = 66.6667 - 5.1190 \times 6$

$= 66.6667 - 30.7143$

$= 35.9524$

Step 3: Write the regression equation.

$\hat{y} = 35.95 + 5.12x$

Step 4: Interpret the coefficients.

Slope (b = 5.12): For each additional hour of study, the predicted exam mark increases by approximately 5.12 marks.

Intercept (a = 35.95): A student who studies 0 hours is predicted to score approximately 35.95 marks. (Note: this interpretation may not be meaningful in practice if no student in the data studied 0 hours — extrapolation beyond the data range is unreliable.)

Step 5: Predict the mark for x = 6 hours.

$\hat{y} = 35.95 + 5.12 \times 6$

$= 35.95 + 30.72$

$= 66.67$

Regression equation: $\hat{y} = 35.95 + 5.12x$. Predicted mark for 6 hours of study: 66.67 marks.

22.4 The Least Squares Principle

The regression line is called the "least squares" line because it is the line that minimises the sum of the squared vertical distances (residuals) between the observed data points and the line. In other words, among all possible straight lines you could draw through the data, the least squares line is the one that makes the total squared prediction error as small as possible.

Residual
$e_i = y_i - \hat{y}_i$ The residual is the difference between the actual value and the predicted value. A positive residual means the actual value is above the line; a negative residual means it is below. The sum of all residuals is always zero: $\Sigma e_i = 0$.

22.5 Important Warnings About Regression

1. Do not extrapolate. The regression equation is only reliable within the range of x-values used to build it. In our example, x ranged from 2 to 10 hours. Predicting the mark for x = 20 hours would give $\hat{y} = 35.95 + 5.12 \times 20 = 138.35$, which is impossible (marks are out of 100). The linear relationship may not hold outside the observed range.

2. Correlation does not imply causation. Even if the regression line fits well, you cannot conclude that studying more causes higher marks. There may be confounding variables (e.g. more motivated students both study more and perform better for other reasons).

3. Check for linearity. The regression model assumes a linear relationship. If the true relationship is curved, the regression line will give poor predictions even if r is moderately high. Always plot the data first (scatter plot) to check for linearity.

22.6 Relationship Between r, r², and b

Connection between slope and correlation
$b = r \cdot (s_y) / (s_x)$ where s_y and s_x are the sample standard deviations of y and x respectively. This shows that the slope and the correlation always have the same sign (since standard deviations are always positive).

r^2 = Explained variation / Total variation = $1 - (\Sigma(y - \hat{y})^2) / (\Sigma(y - \bar{y})^2)$

Exam Tip: STA1610 regression questions typically ask you to: (a) calculate the regression equation from a data table, (b) interpret the slope and intercept in context, (c) make a prediction for a given x-value, and (d) comment on the reliability of the prediction. The calculation is identical to the correlation calculation for the first few steps — you build the same table and compute the same sums. The only difference is that you then calculate a and b instead of r.

Exam Tip: A common STA1610 question asks "Can we use this regression equation to predict y when x = [some value far outside the data range]?" The answer is almost always "No, because this would be extrapolation beyond the range of the data, and the linear relationship may not hold outside that range."

TOOL

Each tool below lists the quantities it needs and the formula it applies, so every calculation can be reproduced by hand or on a calculator during the exam. The interactive version performs the arithmetic for you.

Interactive Calculators

Built-in tools for quick calculations — no external apps needed

Z-Score Calculator

Convert a raw score to a z-score or find the probability.

Inputs: **Raw value (x)**, **Mean (μ)**, **Std Dev (σ)**.

Sample Mean & Standard Deviation Calculator

Enter values separated by commas.

Inputs: **Data values**.

Binomial Probability Calculator

Calculate P(X = x) for a binomial distribution.

Inputs: **n (trials)**, **x (successes)**, **p (prob of success)**.

Poisson Probability Calculator

Calculate P(X = x) for a Poisson distribution.

Inputs: **λ (mean rate)**, **x (occurrences)**.

Confidence Interval for Mean (σ known)

Inputs: **Sample mean ($\bar{x}$)**, **σ**, **n**, **Confidence %**.

Z-Test for Mean (σ known)

Inputs: **Sample mean ($\bar{x}$)**, **Hypothesised μ_0**, **σ**, **n**.

TOOL

Complete Formula Sheet

Every formula you need for the STA1610 exam in one place

Descriptive Statistics

Formula	Expression
Sample Mean	$\bar{x} = \Sigma x / n$ or $\bar{x} = \Sigma fx / n$
Grouped Mean	$\bar{x} \approx (\Sigma f x_m) / n$
Median Position	$(n+1) / 2$
Grouped Median	$L + ((n/2 - F) / (f_m))w$
Grouped Mode	$L + ((d_1) / (d_1+d_2))w$
Range	$x_{max} - x_{min}$
Sample Variance	$s^2 = (\Sigma(x-\bar{x})^2) / (n-1) = (\Sigma x^2 - (\Sigma x)^2/n) / (n-1)$
Sample Std Dev	$s = \sqrt{(s^2)}$
Coeff of Variation	$CV = s / \bar{x} \times 100\%$
IQR	$Q_3 - Q_1$
Quartile Position	Q_k: $(k(n+1)) / 4$
Percentile Position	P_k: $k / 100(n+1)$

Probability

Formula	Expression
Classical Probability	$P(E) = n(E) / n(S)$
Complement	$P(A') = 1 - P(A)$
Addition (general)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Addition (mutually excl.)	$P(A \cup B) = P(A) + P(B)$
Conditional	$P(A \mid B) = P(A \cap B) / P(B)$
Multiplication (general)	$P(A \cap B) = P(A) \cdot P(B \mid A)$
Multiplication (independent)	$P(A \cap B) = P(A) \cdot P(B)$
Bayes' Theorem	$P(A_i \mid B) = (P(A_i)P(B \mid A_i)) / (\Sigma P(A_j)P(B \mid A_j))$
Combinations	$C(n, r) = n! / (r!(n-r)!)$
Permutations	${}^nP_r = n! / ((n-r)!)$

Discrete Distributions

Discrete random variable: $E[X] = \Sigma xP(x)$; $Var(X) = \Sigma x^2P(x) - (E[X])^2$.

Distribution	PMF	Mean	Variance
Binomial	$C(n, x)p^x(1-p)^{n-x}$	np	np(1-p)
Poisson	$(e^{-\lambda}\lambda^x) / x!$	λ	λ

Normal Distribution & Sampling

Formula	Expression
Z-score (individual)	$z = (x - \mu) / \sigma$
Z-score (sample mean)	$z = (\bar{x} - \mu) / (\sigma/\sqrt{(n)})$
Z-score (proportion)	$z = (\hat{p} - p) / (\sqrt{(p(1-p)/n)})$
X from Z	$x = \mu + z\sigma$
Standard Error (mean)	$\sigma_{\bar{x}} = \sigma / \sqrt{(n)}$
Standard Error (prop.)	$\sigma_{\hat{p}} = \sqrt{((p(1-p)) / n)}$

Confidence Intervals

Parameter	CI Formula
Mean (σ known)	$\bar{x} \pm z_{\alpha/2} \cdot \sigma / \sqrt{(n)}$
Mean (σ unknown)	$\bar{x} \pm t_{\alpha/2,n-1} \cdot s / \sqrt{(n)}$
Proportion	$\hat{p} \pm z_{\alpha/2} \cdot \sqrt{((\hat{p}(1-\hat{p})) / n)}$
Sample size (mean)	$n = ((z \cdot \sigma) / E)^2$
Sample size (prop.)	$n = \hat{p}(1-\hat{p})(z / E)^2$

Hypothesis Testing

Test	Test Statistic
Z-test for mean (σ known)	$z = (\bar{x} - \mu_0) / (\sigma/\sqrt{(n)})$
T-test for mean (σ unknown)	$t = (\bar{x} - \mu_0) / (s/\sqrt{(n)})$, df=n-1
Z-test for proportion	$z = (\hat{p} - p_0) / (\sqrt{(p_0(1-p_0)/n)})$
Chi-square	$\chi^2 = \Sigma ((O-E)^2) / E$
Correlation test	$t = (r\sqrt{(n-2)}) / (\sqrt{(1-r^2)})$, df=n-2

Correlation & Regression

Formula	Expression
Correlation r	$r = (n\Sigma xy - \Sigma x \Sigma y) / (\sqrt{([n\Sigma x^2-(\Sigma x)^2][n\Sigma y^2-(\Sigma y)^2])})$
Coeff of Determination	r^2
Regression slope b	$b = (n\Sigma xy - \Sigma x \Sigma y) / (n\Sigma x^2 - (\Sigma x)^2)$
Regression intercept a	$a = \bar{y} - b\bar{x}$
Regression equation	$\hat{y} = a + bx$